Virasoro Constraints Under Projectivization

1 Introduction

1.1 What is being proved, informally

Let Y be a smooth complex projective variety. Gromov–Witten (GW) theory "counts" maps from compact curves into Y; each such map contributes a number that also remembers two pieces of data:

- a homology class $d \in H_2(Y;\mathbb{Z})$ telling us which curve class of Y we are counting;
- at each of the n marked points x_i on the domain curve, a cohomology class $b_i \in H^*(Y;\mathbb{C})$ together with a power $\psi_i^{k_i}$ of the first Chern class of the cotangent line at x_i.

The resulting *descendant integral* is

$$\int_{[\overline{\mathcal{M}}_{g,n}(Y,d)]^{\mathrm{vir}}} \psi_1^{k_1}\,\mathrm{ev}_1^*\,b_1 \cdots \psi_n^{k_n}\,\mathrm{ev}_n^*\,b_n.$$

Stacking *every* such number (all g, d, n, all k_i, all b_i) into a single formal generating series gives the *total descendant potential* $Z_Y(t)$, defined precisely in Section 2.

The *Virasoro constraints* are a sequence of differential equations $L_k^Y Z_Y = 0$, $k \geq -1$, in the sources t. They were proposed by Eguchi–Hori–Xiong and refined by Getzler, and are known for target curves, toric targets, and targets with semisimple quantum cohomology, but remain open in general. This paper proves that they pass from a base to the projectivization of any rank ≥ 2 vector bundle over it. The transfer needs no semisimplicity of quantum cohomology and no positivity or splitness of the bundle.

The precise statement is:

Theorem 1.1. *Let B be a smooth connected projective variety over $\mathbb{C}$, let E be an algebraic vector bundle of rank $r \geq 2$ on B, and set $X = \mathbb{P}_B(E)$. If the ordinary descendant potential of B is annihilated by every Virasoro mode, $L_k^B Z_B = 0$ for all $k \geq -1$, then the same holds for X: $L_k^X Z_X = 0$ for all $k \geq -1$.*

Here Z_Y is the *ordinary* descendant potential of Y (no ancestor classes, no twisting), on the full cohomology $H^*(Y;\mathbb{C})$, with each actual integral curve class kept separate. The conclusion includes every genus, every individual integral curve class, every finite list of descendants, and primitive and odd insertions. Applying the theorem repeatedly covers towers $X \to X' \to \cdots \to B$ of projective bundles over any base satisfying the hypothesis.

1.2 Why this is not obvious

If E splits as $L_1 \oplus \cdots \oplus L_r$, mirror-symmetry arguments prove the transfer (Coates–Givental–Tseng), and if B has semisimple quantum cohomology, semisimplicity transfers to X (Iritani–Koto) and Givental–Teleman reconstruction then gives the constraints on X. Neither assumption is available in general. Non-split bundles and non-semisimple bases require an argument that does not go through mirror symmetry.

Our proof uses a model geometry that sees both B and X as fixed loci, plus a $\mathbb{C}^*$-equivariant shift operator that turns the "change of fixed component" into "change of spectral block." The shift and its spectrum do not depend on semisimplicity. A separate step (Quantum Riemann–Roch) carries the constraints from the local neighborhood of each fixed component to its ordinary theory. The sum over the fiber's spectral cluster assembles the constraints on X.

1.3 The master space and the five ideas

Twist E by a sufficiently negative line bundle on B. This does not change $X = \mathbb{P}_B(E)$, but makes E^* globally generated, hence makes tautological degrees nonnegative on effective curves. Set

$$W = \mathbb{P}_B(E \oplus \mathcal{O}_B).$$

Let $T = \mathbb{C}^*$ act with weight 0 on E and weight 1 on $\mathcal{O}_B$, and let $\lambda \in H^2_T(\mathrm{pt})$ be the equivariant parameter. The fixed scheme is a disjoint union $W^T = B \sqcup X$: the section $[0:1]$ is B, and the section $[1:0]$ is X.

The proof has five steps:

1. **Localize the master ancestors.** The equivariant descendant theory of W factorizes, via the stable-graph sum of virtual localization, into the two fixed-component theories, each twisted by the inverse Euler class of its normal bundle, plus an upper-triangular symplectic transformation $R(z)$: see Proposition 4.1 below. The R and its inverse are explicitly quantizable.

2. **Identify the Iritani shift and its spectrum.** Iritani's equivariant shift operator T_a translates $\lambda \mapsto \lambda + z$. It has a closed formula $T_a = S_W T_d S_W^{-1}$, and on the fiber part of the degree-zero cohomology, its leading term is multiplication by $p + \lambda$ in the quantum algebra of the fiber. Hence the distinct eigenvalue branches are $\lambda + h$, with $h^r(h + \lambda) = y$ and y the fiber class. There are $r + 1$ branches; near $y = 0$, one tends to 0 (the B *branch*), and the other r tend to λ (the X *cluster*).

3. **Isolate spectral blocks.** Contour integrals against the resolvent of T_a produce projectors P_i and logarithms $\mathscr{L}_i$ on each branch. Because the graded operator G_a (which has $\lambda\partial_\lambda$ in it) commutes with T_a, the formula $D_i = P_i G_a - (\lambda/z)\mathscr{L}_i$ has no λ-differentiation. The induced loop operators on blocks are therefore quantizable over the coefficient ring.

4. **Continue around the branch cover.** The cover $h \mapsto h^r(h + \lambda)$ is connected away from its critical values, so monodromy in y transports every equation from the B branch to every other branch. Ancestor coefficients are polynomial in y, and their cotangent powers are bounded independently of fiber degree; consequently the interchange of "formal identity" and "identity of germs" is legal coefficientwise. Adding the r branches of the X cluster gives the constraints on X after going back to $y = 0$.

5. **Fix the scalar normalization.** Quantized transports are only defined projectively. Two commutators, $[L^Y_{-1}, L^Y_1] = -2L^Y_0$ and $[L^Y_0, L^Y_k] = -kL^Y_k$, remove the insertion-independent scalars and give exactly the prescribed constant C_Y in L^Y_0.

Two ingredients of the argument are worth flagging, since they are reusable: (1) subtracting a block logarithm from a differential operator in λ converts a derivative into a loop operator over the coefficient ring; and (2) ancestor bounds make every coefficient test finite, so analytic continuation between branches is legitimate coefficientwise. Neither ingredient is special to projective bundles.

1.4 How to read this paper

Section 2 fixes conventions once and for all: descendant potentials, loop space, ordinary modes, normal ordering, projective constraints, and coefficientwise finiteness. Section 3 constructs the master space, its fixed components, and the pairings between them. Section 4 proves the localization factorization of ancestors on W. Section 5 defines the calibrated grading and the Iritani shift, computes the fiber spectrum, and records the commutation $[G_a, T_a] = 0$. Section 6 constructs the block projectors and logarithms and their modes. Section 7 identifies, via Quantum Riemann–Roch, the block operators on each fixed component with the ordinary Virasoro operators. Section 8 handles monodromy, scalar normalization, and finishes the proof of Theorem 1.1.

Unproved but citable facts are limited to: virtual localization (Graber–Pandharipande), Iritani's equivariant shift theorem, Quantum Riemann–Roch (Coates–Givental), and Givental's upper triangular quantization formula. All other statements are proved in place, with the argument written out.

2 Conventions: descendants, quantization, Virasoro

Throughout the paper, Y is a smooth connected projective variety over $\mathbb{C}$ of dimension d. The two examples to keep in mind are $Y = B$ and $Y = X = \mathbb{P}_B(E)$; every definition in this section applies verbatim to both.

2.1 State space and grading

The state space is the full super-vector space $H_Y = H^*(Y;\mathbb{C})$ with parity given by cohomological degree modulo 2 and Poincaré pairing $\eta_Y(a,b) = (a,b)_Y = \int_Y a \cup b$. Tensor products, permutations, contractions and derivatives follow the Koszul rule; the coevaluation tensor is the categorical inverse of η_Y, without an extra parity sign.

Two even endomorphisms of H_Y organize the Virasoro construction:

$$\mu_Y\big|_{H^{a,b}(Y)} = (a - d/2)\,\mathrm{id}, \qquad R_Y = c_1(TY)\cup(-). \tag{1}$$

Here a is the first Hodge index, so μ_Y uses the first Hodge grading (which is the correct choice for targets with odd cohomology; see Getzler). Poincaré duality gives $\mu_Y^* = -\mu_Y$ and $R_Y^* = R_Y$, and the Hodge type of c_1 gives $[\mu_Y, R_Y] = R_Y$.

2.2 Descendant potentials and Novikov completion

Fix a homogeneous Hodge basis (ϕ_a) with $\phi_0 = 1_Y$. Let t_j^a be formal supercommuting variables with the parity of ϕ_a, $j \geq 0$, and put $t_j = \sum_a t_j^a \phi_a$, $t(z) = \sum_{j\geq 0} t_j z^j$. Define the connected ordinary unreduced descendant potentials and total potential by

$$F_g^Y(t) = \sum_{\substack{\beta \text{ effective} \\ m\geq 0}} \frac{Q^\beta}{m!} \int_{[\overline{\mathcal{M}}_{g,m}(Y,\beta)]^{\mathrm{vir}}} \prod_{i=1}^m \Big(\sum_{j\geq 0} \psi_i^j \, \mathrm{ev}_i^* \, t_j \Big), \qquad Z_Y(t) = \exp \sum_{g\geq 0} \hbar^{g-1} F_g^Y(t). \tag{2}$$

Only stable maps contribute. In particular, ψ_i is the map cotangent class at the i-th mark, not an ancestor class. The exponential records disconnected domains and the empty domain.

Novikov variables remember the actual effective integral class $\beta \in H_2(Y;\mathbb{Z})$, not just its numerical degree: $Q^{\beta_1} Q^{\beta_2} = Q^{\beta_1+\beta_2}$ and $\beta = 0$ corresponds to $Q^0 = 1$. Fix any ample integral class on Y to give the Novikov completion: only finitely many effective β lie below any fixed ample degree. All identities below are read coefficientwise in this completion, at finite order in the descendants, and coefficientwise in $\hbar$ (which is a formal Laurent variable).

2.3 Loop space and ordinary Virasoro modes

Set

$$\mathcal{H}_Y = H_Y((z^{-1})), \quad \Omega_Y(f,g) = \mathrm{Res}_{z=0} \big(f(-z), g(z) \big)_Y \, dz,$$

with polarization $\mathcal{H}_Y = H_Y[z] \oplus z^{-1} H_Y[[z^{-1}]]$. The infinitesimal symplectic operators are

$$\ell_{-1,Y} = z^{-1}, \quad \ell_{0,Y} = z\partial_z + \tfrac{1}{2} + \mu_Y + \frac{R_Y}{z}, \quad \ell_{k,Y} = \ell_{0,Y} (z\ell_{0,Y})^k, \quad k \geq 1. \tag{3}$$

The symplectic identity is exactly the adjoint triple $\mu_Y^* = -\mu_Y$, $R_Y^* = R_Y$, $[\mu_Y, R_Y] = R_Y$ combined with the sign of z in the residue pairing; we check this in the proof of Lemma 8.5 below, where we use it.

2.4 Quantization and Virasoro operators

Write q_j^a for the positive loop coordinates (dual to t_j^a in the standard positive part) and $p_{j,a}$ for the dual negative coordinates. Quantize a quadratic Hamiltonian $\mathcal{Q}_A(f) = \frac{1}{2}\Omega_Y(f, Af)$ by the rule: a qq monomial becomes $q^\top q/(2\hbar)$, a pq monomial becomes the first-order operator with left multiplication before left derivative, and a pp monomial becomes $\hbar$ times the second derivative; all reorderings have their graded signs. Then perform the dilaton translation

$$q(z) = t(z) - z 1_Y. \tag{4}$$

Denote the resulting operator by $\mathrm{op}_{\eta_Y}(A)$.

With these conventions:

$$L_{-1}^Y = \mathrm{op}_{\eta_Y}(\ell_{-1,Y}) = -\frac{\partial}{\partial t_0^0} + \sum_{j\geq 0,a} t_{j+1}^a \frac{\partial}{\partial t_j^a} + \frac{(t_0,t_0)_Y}{2\hbar}, \tag{5}$$

$$L_k^Y = \mathrm{op}_{\eta_Y}(\ell_{k,Y}) \quad (k \neq 0), \qquad L_0^Y = \mathrm{op}_{\eta_Y}(\ell_{0,Y}) + C_Y, \tag{6}$$

$$C_Y = \frac{\chi(Y)}{16} - \frac{1}{4}\,\mathrm{str}(\mu_Y^2).$$

The supertrace counts even minus odd cohomological parity. Positive modes have no scalar correction.

Write $\mathsf{V}(Y)$ for the conjunction of identities

$$L_k^Y Z_Y = 0, \qquad k \geq -1. \tag{7}$$

These identities are read coefficientwise: separately at each genus, at each actual integral class, at each number of marks, and at each monomial in the descendants. The main theorem is $\mathsf{V}(B) \Rightarrow \mathsf{V}(X)$.

2.5 Projective satisfaction and covariance under frame change

Quantization of an infinitesimal symplectic mode is defined only up to an additive scalar. Consequently a change of frame can change the mode equation by a constant. For transfer purposes we track the weaker projective version, then remove the scalar ambiguity at the end.

Definition 2.1 (Projective satisfaction)**.** An even infinitesimal symplectic A is *satisfied projectively* by Z if $Z^{-1}\operatorname{op}_\eta(A)Z$ is independent of the descendants (it may still depend on the Novikov variables, λ, and $\hbar$).

Lemma 2.2 (Projective covariance)**.** *Let M be a symplectic map between paired loop spaces $(\mathcal{H}_Y, \eta_Y)$ and $(\mathcal{H}_{Y'}, \eta_{Y'})$, with both M and M^{-1} admitting quantized actions $U_M, U_{M^{-1}}$ in the coefficientwise completion. If A and MAM^{-1} are infinitesimal symplectic and quantizable, then*

$$U_M \operatorname{op}_{\eta_Y}(A) U_M^{-1} = \operatorname{op}_{\eta_{Y'}}(MAM^{-1}) + \textit{scalar}. \tag{8}$$

In particular, A is satisfied projectively by Z iff MAM^{-1} is satisfied projectively by $U_M Z$; multiplying either potential by an invertible scalar does not affect the projective condition.

Proof. The commutator of two normally ordered quadratic operators is the quantization of their Hessian/Hamiltonian bracket plus a scalar double contraction, which in super Darboux coordinates is a supertrace. Exponentiating this identity gives (8); equivalently this is Givental's projective-quantization formalism. Neither U_M nor MAM^{-1} differentiates $\hbar$ or the parameters, so the projective condition is preserved. □

2.6 Coefficientwise finiteness

Some of the mode matrices defined below have infinitely many positive powers of z, or finite-order $z\partial_z$ derivatives. The following lemma collects the formal facts we need.

Lemma 2.3 (Coefficientwise finiteness)**.** *Suppose a potential has finitely many descendant coefficients at each fixed curve class, $\hbar$-power, and total descendant degree, and has $\hbar$-powers bounded below at each fixed curve class and total degree. Let A be an even infinitesimal symplectic mode with a finite z-lower-bound and finite order in $z\partial_z$ at every filtration step. Then $\operatorname{op}_\eta(A)$ acts coefficientwise on the potential. The same holds with additional formal filtration variables; and quantized actions U_M of transformations equal to the identity modulo positive Novikov filtration are defined by their formal exponential and inverse series.*

Proof. At fixed filtration, the pp part differentiates only finitely many coefficients. In the pq part a bounded differentiated index and a lower bound on loop powers leaves only finitely many

multiplied indices. In the qq part the order in $z\partial_z$ is finite. Euler differentiation $z\partial_z$ multiplies mode coefficients by polynomials in their indices and preserves these bounds. At fixed positive filtration, only finitely many terms of an exponential can occur. □

Ordinary descendant potentials satisfy the required support: for fixed genus, class and number of marks, high enough powers of ψ_i vanish on the finite-dimensional stable-map space. Ancestor potentials satisfy a stronger bound from the stable-curve moduli space (Lemma 8.1 below).

Categorical inverse and signs. All splitting and cotangent-class identities used later are tensor identities built from the categorical inverse of η_Y. Their standard marked-point proofs therefore apply to full cohomology with these signs. Quantum Riemann–Roch and the two-mark shift identities likewise use arbitrary evaluation classes and this diagonal.

3 The master space and its fixed theories

We put B and $X = \mathbb{P}_B(E)$ into one smooth projective variety W with a torus action, so that both are fixed loci. Virtual localization will then express the theory of W in terms of the theories of B and X. This section sets up the geometry, curve labels, pairings, and the ancestor formalism.

3.1 Geometry and integral curve labels

Tensoring E by a line bundle on B does not change $X = \mathbb{P}_B(E)$. Choose such a twist so that E^* is globally generated, and (abusing notation) continue to write E for the twisted bundle. Set

$$X = \mathbb{P}_B(E), \qquad W = \mathbb{P}_B(E \oplus \mathcal{O}_B).$$

Let $T = \mathbb{C}^*$ act with weight zero on E and weight one on $\mathcal{O}_B$. Write $\lambda \in H^2_T(\mathrm{pt})$ for the weight-one representation's class. The fixed scheme is $W^T = X \sqcup B$, where the second component is the section corresponding to the summand $\mathcal{O}_B$.

Write π for the projective-bundle projection of either X or W to B, and put $p = c_1^T(\mathcal{O}_W(1))$. The normal bundles, their ranks, weights, and the integers W_F, j_F that will appear in the shift formula below are:

F	N_F	$n_F = \operatorname{rk} N_F$	w_F	$W_F = n_F w_F$	j_F
X	$\mathcal{O}_X(1)$	1	$+1$	$+1$	0
B	E	r	-1	$-r$	1

(9)

The restrictions of the equivariant divisor are

$$p|_X = c_1(\mathcal{O}_X(1)), \qquad p|_B = -\lambda.$$

Here w_F is the common $\mathbb{C}^*$-weight on the normal fibers. Both descriptions come from the standard relative tangent sequence: at a line $L \subset V$ one has $T_{W/B} \cong \operatorname{Hom}(L, V/L)$.

Lemma 3.1. *For $P = X$ or $P = W$, the map*

$$H_2(P;\mathbb{Z}) \to H_2(B;\mathbb{Z}) \oplus \mathbb{Z}, \quad d \mapsto \Big(\pi_* d, \int_d c_1(\mathcal{O}_P(1))\Big) \tag{10}$$

is an isomorphism, including torsion. An effective class has effective base pushforward and nonnegative tautological degree.

Proof. The structure group of a projective bundle acts trivially on the integral homology of the fiber, so the Serre spectral sequence in total degree two is $H_2(B;\mathbb{Z}) \oplus H_2(\mathbb{P}^{s-1};\mathbb{Z})$ with $s \geq 2$ the rank. The fiber generator survives integrally because $c_1(\mathcal{O}_P(1))$ pairs to one with the fiber line: its image in total homology cannot be zero or a nontrivial multiple. Thus we have an exact sequence

$$0 \to \mathbb{Z}[\text{fiber line}] \to H_2(P;\mathbb{Z}) \xrightarrow{\pi_*} H_2(B;\mathbb{Z}) \to 0,$$

and pairing with $c_1(\mathcal{O}_P(1))$ provides a retraction on the first term, giving (10). Finally, E^* and $E^* \oplus \mathcal{O}_B$ are globally generated, so the tautological line bundle is globally generated and has nonnegative degree on effective curves; proper pushforward preserves effectivity. □

Henceforth we write $Q^\beta y^l$ for the actual class (β, l). Under $X \hookrightarrow W$ the pair (β, l) is unchanged; under $B \hookrightarrow W$ a class β becomes $(\beta, 0)$.

Completion. Fix an ample integral class H on B; for M large enough, $c_1(\mathcal{O}_P(1)) + M\pi^*H$ is ample on both projective bundles. We complete in the degree $l + M\int_\beta H$. Only finitely many effective labels occur below each fixed degree. Unless otherwise noted, "positive Novikov filtration" means positive degree in this completion (so it includes positive fiber degree even for zero base class).

3.2 Equivariant pairings and inverse-Euler twists

The equivariant projective-bundle formula makes $H_T^*(W)$ free over $\mathbb{C}[\lambda]$ with basis $\pi^*\phi_a \cdot p^j$, $0 \leq j \leq r$, for a homogeneous Hodge basis (ϕ_a) of H_B. Equivariant integration pairs this basis perfectly over $\mathbb{C}[\lambda]$: fiberwise integration gives an antitriangular matrix whose antidiagonal blocks are the ordinary Poincaré pairing of B.

After extending scalars to $\mathbb{C}(\lambda)$, restriction to the fixed locus is an isomorphism of paired spaces

$$(H_T^*(W), \eta_W) \otimes_{\mathbb{C}[\lambda]} \mathbb{C}(\lambda) \cong \bigoplus_{F=B,X} (H_F \otimes \mathbb{C}(\lambda), \eta_F^t), \qquad \eta_F^t(a,b) = \int_F \frac{a \cup b}{e_T(N_F)}. \tag{11}$$

The unit restricts to the sum of the two fixed units. The Euler classes are invertible over $\mathbb{C}(\lambda)$ because both normal weights are nonzero.

The corresponding *inverse-Euler-twisted* Gromov–Witten theory on F inserts

$$e_T(R^\bullet \rho_* f^* N_F)^{-1} \tag{12}$$

in each stable-map integral, extended multiplicatively to the indicated K-theory class. Here ρ and f are the universal curve and universal map. We use subscripts F, tw for its potentials and fundamental solutions. Its degree-zero three-point pairing is exactly η_F^t.

Hodge bidegrees. Give p and λ Hodge bidegree $(1,1)$. In a homogeneous polynomial basis, μ acts on basis vectors by the first Hodge degree minus $\dim_{\mathbb{C}}/2$, and differentiation in λ records the degree of equivariant coefficients. Virtual dimension identities can be expressed in this first degree: algebraicity of evaluation maps and the virtual class forces the sum of Hodge-degree differences of insertions to be zero in a nonzero invariant. This remains true equivariantly (check on finite-dimensional approximations or after fixed-locus integration). This is a separate grading from the cohomological degree that later bounds the fiber degrees.

3.3 Ancestors and the fundamental solution

We now fix notation for ancestors, applicable to an ordinary theory, an equivariant theory, or one of the twisted fixed theories. Denote its pairing by η, its Novikov monomial for class d by q^d, its stable-map integrals by brackets. Let u be a formal even primary background. For distinguished insertions $a_1, \dots, a_m$ set

$$\langle a_1, \dots, a_m \rangle_{g,u} = \sum_{\substack{d \text{ eff.} \\ n \geq 0}} \frac{\mathsf{q}^d}{n!} \langle a_1, \dots, a_m, \underbrace{u, \dots, u}_{n} \rangle_{g,m+n,d}.$$

Here d runs over effective classes and only stable-map terms are included. Descendants in these brackets use map cotangent classes.

For $2g - 2 + m > 0$ there is a stabilization map

$$\mathrm{st}_{g,m} : \overline{\mathcal{M}}_{g,m+n}(Y, d) \to \overline{\mathcal{M}}_{g,m}$$

forgetting the map and the n background marks. Set $\bar{\psi}_i = \mathrm{st}_{g,m}^* \psi_i$, $1 \leq i \leq m$. The ancestor potential at background u is

$$\overline{F}_g(u;t) = \sum_{\substack{d \text{ eff.}, \ m,n \geq 0 \\ 2g-2+m>0}} \frac{\mathsf{q}^d}{m!n!} \Big\langle \prod_{i=1}^{m} \Big(\sum_{j \geq 0} \bar{\psi}_i^j \, \mathrm{ev}_i^* \, t_j \Big) \prod_{a=m+1}^{m+n} \mathrm{ev}_a^* \, u \Big\rangle_{g,m+n,d}, \quad \mathcal{A}(u;t) = \exp \sum_{g \geq 0} \hbar^{g-1} \overline{F}_g(u;t). \tag{13}$$

The bracket denotes integration of the indicated product, with the twist when appropriate. Terms with curve-unstable distinguished data are omitted even when the stable-map space itself is nonempty; in particular the genus-one term with no distinguished marks is omitted. Ancestors come from $\overline{\mathcal{M}}_{g,m}$, so a total power above $3g - 3 + m$ vanishes.

Definition 3.2 (Fundamental solution)**.** The fundamental solution $S(u, z) = \mathrm{id} + O(z^{-1})$ is defined by

$$\eta(b, S(u,z)a) = \eta(b, a) + \Big\langle b, \frac{a}{z - \psi} \Big\rangle_{0,u}, \qquad \frac{1}{z - \psi} = \sum_{j \geq 0} \psi^j z^{-j-1}. \tag{14}$$

The pairing term supplies the unstable two-point contribution.

The genus-zero splitting and cotangent-class identities give

$$S(u, -z)^* S(u, z) = \mathrm{id}, \qquad z \partial_v S(u, z) = (v \star_u) S(u, z), \tag{15}$$

where $\star_u$ is the quantum product and ∂_v differentiates the primary background. Recall that the genus-zero cone $\mathcal{L}$ is the formal Lagrangian graph of dF_0 in Darboux coordinates (q, p) with $q = t - z1$. Its point with descendant input ϵ has positive coordinate $\epsilon - z1$, and the tangent space there is the graph of the Hessian of F_0. The string, dilaton and topological recursion identities make this tangent space tangent along $zT \subset \mathcal{L}$; at primary coordinate u the tangent space is $S(u, z)^{-1} \mathcal{H}_+$ (Givental). The ancestor–descendant correspondence in our group-action convention is

$$\mathcal{A}(u) = c(u) U_{S(u)} Z, \tag{16}$$

with $c(u)$ independent of the ancestor variables. Both formulas follow from the splitting identity for $\psi_i - \bar{\psi}_i$ and its genus-zero specializations (Kontsevich–Manin, Getzler, Givental), in the super conventions of Section 2.

We use (16) only on the fixed theories, with u of positive total Novikov filtration. On such u, $S(u,z) - \mathrm{id}$ has positive filtration, and every Novikov coefficient has finitely many negative z-powers: finitely many background marks at that coefficient, and F's cotangent classes are nilpotent. The quantized action is then covered by Lemma 2.3.

For the master space we abbreviate $S_W = S_W(0,z)$ and $\mathcal{A}_W = \mathcal{A}_W(0)$. Its loop-variable rationality and its quantized action against the fixed theories will be established directly by localization in the next section.

4 Ancestor localization for the master space

The torus action on W splits its Gromov–Witten theory into contributions from B and X. This section proves the resulting factorization of the master-space ancestor potential. We abbreviate

$$H^{\mathrm{fix}} = H_B \oplus H_X, \qquad \eta^{\mathrm{fix}} = \eta_B^t \oplus \eta_X^t.$$

Restriction identifies this paired space with $H_T^*(W)$ after inverting λ. All series use full curve labels $Q^\beta y^l$.

Proposition 4.1 (Ancestor localization)**.** *There are classes $u_F \in H_F$ of positive Novikov filtration for $F = B, X$, and a symplectic power series $R(z) \in \mathrm{End}(H^{\mathrm{fix}})[[z]]$ with $R - \mathrm{id}$ of positive Novikov filtration, such that, writing $S_f(z) = S_{F,\mathrm{tw}}(u_F, z)$ and $S_{\mathrm{bl}}(z) = \bigoplus_F S_f(z)$,*

$$S_W(z) = R(z) S_{\mathrm{bl}}(z), \tag{17}$$

$$\mathcal{A}_W(0) = c\, U_R \prod_{F=B,X} \mathcal{A}_{F,\mathrm{tw}}(u_F), \tag{18}$$

with c an invertible scalar independent of ancestor variables. For every curve coefficient, $S_W(z)$ is the expansion of a rational function of z (at $z = 0$ the identity is in Laurent series). Both R and its inverse admit quantized actions in the coefficientwise completion used here.

Source of the argument. The same master-space geometry underlies Fan's reconstruction of projective-bundle invariants; for ancestors we adapt Coates–Givental–Tseng's localization, with an additional gluing argument required because the orbit directions here vary over the positive-dimensional variety X.

4.1 Moving components and their gluing

A *moving leg* is an irreducible component of a torus-fixed stable map whose image is not contained in $W^T = B \sqcup X$. Connected pieces mapping into B or X are kept as stable-map factors for that component (including their internal boundary strata).

Lemma 4.2 (Leg geometry)**.** *A moving leg has degree $m \geq 1$ on a line joining $x \in X$ to $b = \pi(x) \in B$. Its map from $\mathbb{P}^1$ is totally ramified at the two endpoints, and its markings and nodes lie at these endpoints. The leg moduli is the smooth proper μ_m-gerbe $q_m : \mathfrak{L}_m \to X$ of mth roots of $\mathcal{O}_X(-1)$, with endpoint evaluation maps q_m and $\pi \circ q_m$.*

At an endpoint in F, the tangent line of the leg has equivariant first Chern class

$$a = \frac{w_F \lambda}{m} + \nu,$$

with ν an ordinary degree-two class on $\mathfrak{L}_m$. Every node involving a moving leg has nonzero smoothing character.

Proof. After a finite cover of the torus, the action lifts to the domain of a fixed stable map. A nontrivial $\mathbb{C}^*$-action on a complete irreducible curve is trivial on its normalization; since T acts trivially on B the projection of a moving component is constant. In a fiber $\mathbb{P}(E_b \oplus \mathbb{C})$ the closure of any nonfixed orbit is the line joining $[0:1]$ to a unique $[v:0] \in \mathbb{P}(E_b)$. An equivariant finite map to this closure is, at the two endpoints, $t \mapsto t^m$; a self-node would identify distinct fixed points, so the moving component is smooth and its special points all lie at the endpoints.

Varying the endpoint $[v]$ gives X as the parameter space of orbit lines. With $q_m : \mathfrak{L}_m \to X$ the root gerbe of $\mathcal{O}_X(-1)$ and M the universal mth-root line, the universal cover

$$\mathbb{P}(M \oplus \mathcal{O}) \to \mathbb{P}(q_m^* \mathcal{O}_X(-1) \oplus \mathcal{O}) \to W, \qquad [s:t] \mapsto [s^m : t^m],$$

is locally the moving leg, and its automorphisms are exactly the deck group μ_m. This identifies the leg moduli. It is smooth and proper over the projective variety X, and its ordinary class coincides with its fixed virtual class: on a given fiber line L,

$$T_{W/B}|_L \cong \mathcal{O}(2) \oplus \mathcal{O}(1)^{\oplus(r-1)}, \quad f^*\pi^* T_B \cong T_{B,b} \otimes \mathcal{O}_{\mathbb{P}^1},$$

so $H^1(\mathbb{P}^1, f^*T_W) = 0$; the fixed part of the ambient deformation space is exactly $T_{\mathfrak{L}_m}$.

The universal cover gives tangent classes $a_X = (\lambda + q_m^* p|_X)/m$ at X and $a_B = -a_X$ at B, with scalar characters $+\lambda/m$ at X and $-\lambda/m$ at B. A node joining a leg to a fixed-map factor has the corresponding nonzero character. Two legs at a node approach the same fixed component from opposite signs, so the smoothing character is $w_F \lambda(1/m + 1/m') \neq 0$. □

4.2 Gluing and virtual localization

Localization is applied as follows. Distinguish all cut branches at nodes, then divide by graph automorphisms. Since every smoothing character at a node cut from a moving leg is nonzero, fixed deformations cannot smooth those nodes; fixed loci are described by matching leg endpoints and stable-map pieces to B or X along evaluation maps. This description holds in families: degree zero over B forces the projection of a rational family to factor through its base, and the local monomial form applies after a trivialization.

Obstruction theory matching. Normalize pieces C_α at the cut nodes q. The map-deformation complexes fit into the compatible triangle

$$R\Gamma(C, f^*T_W) \to \bigoplus_\alpha R\Gamma(C_\alpha, f_\alpha^* T_W) \to \bigoplus_q T_{W, f(q)}.$$

Passing to absolute map deformations adds pointed-domain deformation and automorphism complexes, and releasing a cut node adds its smoothing line $T_{q,+} \otimes T_{q,-}$. The ambient virtual tangent class of the fixed gluing locus is thus

$$[\mathbb{T}_\Gamma^{\mathrm{vir}}] = \sum_\alpha [\mathbb{T}_\alpha^{\mathrm{vir}}] - \sum_q [\mathrm{ev}_q^* T_W] + \sum_q [T_{q,+} \otimes T_{q,-}]. \tag{19}$$

The identity comes from the displayed triangle, so the fixed part gives the diagonal-matched obstruction theory, not merely an equality of virtual ranks. On a fixed-map piece the fixed part

is the ordinary theory in F; on a leg it is $T_{\mathfrak{L}_m}$. The matching terms are T_F; every smoothing line is moving. The fixed virtual class is the diagonal virtual pullback of these classes, including when two legs meet or a graph has a cycle.

Cutting a leg off a fixed-map vertex contributes the normal matching numerator $e_T(N_F)$ and the smoothing denominator

$$\frac{1}{a-\psi}, \tag{20}$$

with ψ the cotangent class at the vertex. The numerator exactly turns the contraction into the inverse-Euler-twisted η_F^t. All remaining leg data are cohomology correspondences independent of the vertex descendants. For an *external* marked point at the leg endpoint, there is no smoothing denominator; its cotangent class is $-a$, and contraction with η_F^t again cancels the matching numerator.

4.3 Nilpotent denominators and dummy derivatives

The ordinary class ν in a need not descend to F, so we expand in $\mathfrak{L}_m$ before pushing forward. If $a = a_0 + \nu$ with $a_0 = w_F\lambda/m \neq 0$, then

$$\frac{1}{a-s} = \sum_{j\geq 0} \frac{(-\nu)^j}{(a_0-s)^{j+1}} \tag{21}$$

(finite sum). Let C be a correspondence coefficient on a cut piece with endpoint map e to F, and put $h_j = e_*(\nu^j C)$ (with virtual and Euler factors included). For any flag function K, define

$$K(a)h := \sum_{j\geq 0} \frac{1}{j!}\, \partial_\alpha^j K(\alpha)\big|_{\alpha=a_0}\, h_j, \tag{22}$$

a *dummy* scalar α: the derivative acts only on K. For example $S_f(a)h/(a+z)$ means (22) with $K(\alpha) = S_f(\alpha)/(\alpha+z)$. At several flags use joint moments and independent dummy scalars. A pure ordinary cohomology vector is the special case with only h_0.

Coinvariance and finiteness. All graph sums below are coefficientwise finite: each moving leg has positive tautological degree, so its multiplicity and the number of legs is bounded at fixed ample degree; there are finitely many splittings into effective full curve labels; stability bounds remaining degree zero pieces once genus and distinguished markings are fixed.

4.4 The genus-zero factor R

Fix $F \in \{B, X\}$. An *end* at F is a genus-zero unmarked tree attached to a fixed-map vertex by a moving leg, with the vertex itself omitted. Its contribution, including the $(a-s)^{-1}$ denominator at the attachment, is a cohomology-valued power series $\epsilon_F(s)$ of positive Novikov filtration. A *tail* has one external primary insertion b and contributes $h_b/(a-s)$; the moments h_b absorb curve monomials, stack and symmetry factors, and the linear dependence on b.

End denominators have infinite Taylor expansions; in identities involving these backgrounds we use the completion $\widehat{\mathcal{H}}_+ = H_F[[z]]$ of the positive space, with localized Novikov coefficients, and $[\,\cdot\,]_+$ the nonnegative part. The usual cone identities extend coefficientwise (positive filtration bounds the number of insertions; nilpotence of cotangent classes bounds indices).

Let u_F be the primary parameter of the tangent space to the genus-zero cone of the F,tw theory at background ϵ_F; equivalently it is determined by

$$x_F(z) := [S_f(z)(\epsilon_F(z) - u_F)]_+ \in z\widehat{\mathcal{H}}_+, \quad S_f(z) = S_{F,\mathrm{tw}}(u_F, z). \tag{23}$$

Applying S_f makes the cone point belong to $z\widehat{\mathcal{H}}_+$, and $[S_f(z)(-z+u_F)]_+ = -z$; the linearization of the constant term is $-\mathrm{id}$ at zero Novikov degree, so u_F exists uniquely and has positive filtration.

The tangent space being the Hessian graph implies that its two-point function depends on ϵ_F only through u_F. The genus-zero two-point identity (dimension-counted from topological recursion) is

$$\frac{\eta_F^t(b,c)}{w+z} + \Big\langle \frac{b}{w-\psi}, \frac{c}{z-\psi} \Big\rangle_{0,\epsilon_F}^{F,\mathrm{tw}} = \frac{\eta_F^t(S_f(w)b, S_f(z)c)}{w+z}. \tag{24}$$

It is rational in w, z at each curve coefficient.

Localize the master matrix element $\eta(b, S_W(z)c)$ for c supported on F. If the input lies on a fixed-map vertex, the unmarked outgoing trees provide the background ϵ_F; the insertion b is either on that vertex ($b|_F$) or on a tail with attachment $h_b/(a-\psi)$. If b is a marked leg endpoint, the endpoint rule supplies the metric term in (24), with denominator $a+z$. Taking the w^{-1} coefficient of (24) handles a primary slot, and we get

$$\eta(b, S_W(z)c) = \eta_F^t(b|_F, S_f(z)c) + \sum_{\text{tails}} \frac{\eta_F^t(S_f(a)h_b, S_f(z)c)}{a+z}, \tag{25}$$

giving

$$(R(z)^* b)_F = b|_F + \sum_{\text{tails}} \frac{S_f(a)h_b}{a+z} \tag{26}$$

as a power series at $z=0$. Consequently R is upper triangular and

$$R(-z)^* R(z) = \mathrm{id}, \tag{27}$$

since both S_W and S_{bl} are symplectic and their symbolic identities are rational at each curve coefficient. Every nonidentity term in (26) has a moving leg, so $R - \mathrm{id}$ has positive Novikov filtration.

Translation and bridge. The one-point function is $J_W(-z) = -zS_W(z)^{-1}1$. The string equation identifies it as

$$J_W(-z) = -z1 + \sum_{\beta,l} Q^\beta y^l (\mathrm{ev}_1)_* \frac{[\overline{\mathcal{M}}_{0,1}(W,(\beta,l))]^{\mathrm{vir}}}{-z-\psi_1},$$

where only stable one-marked terms occur. Its regular part on F is $-z + \epsilon_F(z)$ (a marked leg endpoint contributes an end; markings on the fixed vertex contribute only negative powers). Hence $\epsilon_F(z) = z + [J_W(-z)|_F]_+$. Using S_f's nonpositive powers, $[S_f(z)z]_+ = z + u_F$, and the factorized identity gives

$$\bigoplus_F x_F(z) = z(1 - R(z)^{-1}1). \tag{28}$$

A *bridge* is an unmarked genus-zero tree with two fixed-map attachments and moving legs at both; its contribution, dressed by $S_f(a), S_{f'}(a')$ on the two correspondence slots, is $D(s, s')$. Localizing the master two-point function and using 'path determines unique bridge' gives

$$\frac{S_W(w)^* S_W(z)}{w+z} = S_{\mathrm{bl}}(w)^* \Big(\frac{\mathrm{id}}{w+z} + D(-w,-z) \Big) S_{\mathrm{bl}}(z), \tag{29}$$

so, with (17) and (27),

$$D(-w,-z) = \frac{R(w)^* R(z) - \mathrm{id}}{w+z}. \tag{30}$$

The numerator vanishes at $w = -z$, so the right side is a power series at $(w, z) = (0, 0)$, as is the dressed bridge expression; the identity is first proved for symbols and then expanded as a Taylor series.

4.5 From descendants at flags to ancestors

Two ancestor identities convert surviving vertices to ancestor theory. For a fixed target with pairing η^t, let $\mathcal{A}(\epsilon; t)$ denote the mixed potential with distinguished descendants $t(\bar{\psi})$ and additional insertions $\epsilon(\psi)$; the ancestors forget the additional marks. Parameter ϵ of positive filtration; u and $x(z)$ as in (23).

Lemma 4.3 (Vertex conversion)**.** *In summed vertex correlators with background ϵ, an insertion at a distinguished flag transforms as*

$$\frac{h}{a-\psi} \longmapsto \frac{S(u,a)h}{a-\bar{\psi}}, \tag{31}$$

also with additional ancestor-class powers at the flag and with further distinguished flags (all flag evaluation classes in h before applying $S(u,a)$). Moreover, up to a scalar independent of t,

$$\mathcal{A}(\epsilon; t) = \mathcal{A}(u; t + x). \tag{32}$$

Proof. These are the ancestor identities of Getzler and Kontsevich–Manin, in the form used by Coates–Givental–Tseng; we spell out the arguments because the background has descendants.

The identity $\frac{1}{a-\psi} = \frac{1}{a-\bar{\psi}} + \frac{\psi-\bar{\psi}}{(a-\psi)(a-\bar{\psi})}$ reduces (31) to the splitting divisor of $\psi - \bar{\psi}$. That divisor separates a genus-zero twig at the mark, a subset of background marks, and an attaching node. Splitting the virtual class contracts the twig's two-point function with the rest of the vertex. Summing all twigs plus the direct term gives $S(u,a)h$ by the primary-slot instance of (24). Ancestor classes at the original markings are pulled back from the stabilized curve, so they do not enter the twig calculation. For a tangent class with nilpotent part, apply the scalar identity then the finite dummy derivatives of (22).

For (32), write $\epsilon = u + (\epsilon - u)$ and expand multilinearly. Forget the primary u marks alone; the splitting argument converts the remaining descendants to $x(\bar{\psi}) = [S(u,z)(\epsilon(z) - u)]_+|_{z=\bar{\psi}}$. The original ancestors still forget the x marks; the new ones retain them. Their cotangent-class difference has a boundary divisor with a rational tail carrying one original mark and $k \geq 1$ marks to be forgotten; its stable-curve factor is $\overline{\mathcal{M}}_{0,k+2}$ of dimension $k-1$; every x insertion has positive ancestor power (since $x \in z\widehat{\mathcal{H}}_+$), so the product restricts to at least degree k and vanishes. Applying this at every distinguished flag replaces original ancestors by those retaining all x marks; multilinearity gives (32). Terms that only become stable after adding x contribute only a scalar (in genus zero, positive ancestor powers exceed the stable-curve dimension; in genus one, no original distinguished mark leaves a scalar). □

4.6 Assembly

Fix a stable map contributing to a stable ancestor invariant of W. Forget the map and stabilize the distinguished curve. Every moving leg disappears (it has at most two special points). Prune unmarked rational trees and suppress rational chains with two remaining flags. Surviving fixed-map factors are kept with their own stabilization and internal boundary. Removed pieces between surviving vertices are *bridges*; removed pieces ending in a distinguished marking are *tails*; all others are *ends*.

Ends give the background ϵ_F. Lemma 4.3 converts bridge and tail denominators to ancestor denominators and inserts the factors $S_f(a)$. Bridges become exactly $D(\bar\psi, \bar\psi')$. Direct markings together with all tails become, using (26),

$$R(-z)^* t(z) = R(z)^{-1} t(z), \qquad z = \bar\psi.$$

Finally (32) replaces ϵ by u_F and adds x_F. Combined with (28) the vertex substitution is

$$t(z) \longmapsto R(z)^{-1} t(z) + z(1 - R(z)^{-1}1). \tag{33}$$

Quantize this substitution. If $D(s, s') = \sum_{i,j\geq 0} D_{ij} s^i (s')^j$, the upper-triangular contraction operator is $\hbar/2$-times the second derivative with coefficients D_{ij}, using the categorical inverse of η^{tw}. Exponentiating it on the product of fixed ancestors and applying (33) is exactly U_R: the kernel is (30), and the signs $-w, -z$ come from the negative Darboux modes $(-z)^{-i-1}$. The graph sum reproduces this: labeling flags first and then dividing by permutations produces the same factorials and graph automorphisms; a bridge or self-bridge contributes one factor of $\hbar$; super signs and permutations are exactly those of the tensor rules.

Components without distinguished markings may multiply $\mathcal{A}_W$ by an invertible scalar; this is the c in (18). The invertibility of U_R is coefficientwise: $R - \mathrm{id}$ has positive filtration so its inverse and logarithm are defined degree by degree; at each fixed coefficient the number of contributing filtration-positive factors is finite; ancestor powers are bounded by $3g-3+n$; fixed descendant identities are bounded by fixed-target stable-map dimensions. This completes the proof of Proposition 4.1.

5 Grading and the Iritani shift

We now build two commuting operators on the equivariant quantum cohomology of W: the calibrated grading G_a, and Iritani's shift T_a, which translates $\lambda \mapsto \lambda + z$. The T_a admits a simple closed formula on each fixed component, and its spectrum on fiber classes has $r+1$ branches that will organize the transfer.

Throughout $\star$ means the small equivariant quantum product of W at primary background zero. Put

$$c = c_1^T(TW), \qquad \kappa = c_1(TB) + c_1(E).$$

The equivariant projective-bundle formula gives

$$c = (r+1)p + \pi^*\kappa + \lambda, \qquad \int_{(\beta,l)} c_1(TW) = (r+1)l + \int_\beta \kappa. \tag{34}$$

Choose a homogeneous Hodge basis of $H^*(B)$ and the resulting polynomial basis $\pi^* b\, p^j$ $(0 \leq j \leq r)$ of $H_T^*(W)$ over $\mathbb{C}[\lambda]$. In this basis, μ acts by the first Hodge degree minus $\dim_{\mathbb{C}}(W)/2$, and does not differentiate the λ-coefficient.

5.1 The calibrated grading

Define

$$G_d = z\partial_z + \lambda\partial_\lambda + \tfrac{1}{2} + \mu + \frac{c\cup}{z}, \quad G_a = z\partial_z + \lambda\partial_\lambda + \tfrac{1}{2} + \mu + \frac{c\star}{z}. \tag{35}$$

The subscripts distinguish the constant (descendant) frame from the quantum (ancestor) frame. Both differentiate λ and hold Novikov variables fixed.

Lemma 5.1. *With $S_W = S_W(0,z)$ in the conventions of Section 2,*

$$G_a = S_W G_d S_W^{-1}.$$

Proof. Let D_c be the derivation of the Novikov ring sending $Q^\beta y^l$ to $((r+1)l + \int_\beta \kappa)Q^\beta y^l$. Homogeneity of the two-point invariants defining S_W says

$$(z\partial_z + \lambda\partial_\lambda + D_c)S_W = S_W\mu - \mu S_W. \tag{36}$$

This uses the first Hodge degree: evaluation maps, virtual class, and cotangent classes are algebraic, so their push-pull has the Hodge bidegree prescribed by virtual dimension. Concretely, if the input and output basis vectors have first Hodge degrees h_j, h_i, the coefficient of $Q^\beta y^l z^{-k-1}$ in S_W's corresponding entry has λ-degree $h_j + k + 1 - h_i - ((r+1)l + \int_\beta \kappa)$, since λ and z are both bidegree $(1,1)$. This is (36). It applies equally to primitive and odd inputs.

The divisor equation gives

$$zD_c S_W = (c\star)S_W - S_W(c\cup). \tag{37}$$

Substituting into (36) gives $G_a S_W = S_W G_d$. The identities first hold in the $z=\infty$ expansion; by coefficientwise rationality of S_W (from localization) they also hold in the $z=0$ Laurent expansion. □

5.2 The Iritani shift

Iritani's input. Let Y be a smooth projective variety with a $\mathbb{C}^*$-action, form its associated bundle $\widehat{Y} \to \mathbb{P}^1$, and let $\sigma_{\min}$ be the section class of the fixed component with positive normal weights. The operator defined by two-fiber section invariants, with monomials $Q^{\widehat{d}-\sigma_{\min}}$, is an unlocalized equivariant cohomology operator. After reversing the sign of Iritani's loop variable, it translates $\lambda \mapsto \lambda + z$ and satisfies

$$T_a = S_Y T_d S_Y^{-1}, \qquad T_d\big|_F = Q^{\sigma_F - \sigma_{\min}} \prod_{m,\,x} \frac{\prod_{a=-\infty}^{0}(x + m\lambda - az)}{\prod_{a=-\infty}^{-m}(x + m\lambda - az)} e^{z\partial_\lambda}. \tag{38}$$

Here x runs through the ordinary Chern roots of the weight-m normal subbundle of F; each quotient has only finitely many remaining factors. This is the projective specialization of Iritani's shift theorem; the source's seminegativity hypothesis is automatic for a complete target, and its calibration is related to ours by $S_Y(z) = M(-z)^{-1}$.

Associated geometry. For the weight-one action on $\mathcal{O}_B$, the associated space is

$$\widehat{W} = \mathbb{P}_{B\times\mathbb{P}^1}\Big(\mathrm{pr}_B^*\, E \oplus \mathrm{pr}_{\mathbb{P}^1}^*\, \mathcal{O}(-1)\Big) \cong W \times (\mathbb{C}^2 \setminus \{0\})/\mathbb{C}^*, \tag{39}$$

where $s\cdot(w, v_1, v_2) = (s\cdot w, s^{-1}v_1, s^{-1}v_2)$. An additional torus scales v_2; its equivariant parameter is Iritani's loop variable before sign reversal. The fiber over $[1:0]$ carries the original action and the fiber over $[0:1]$ carries the action composed with the additional torus; identifying their cohomologies via the induced change of torus variables is the translation in (38).

Write ζ for that parameter before sign reversal, and put $\rho_0(t,u)w = t\cdot w$, $\rho_1(t,u)w = (tu)\cdot w$. The change of torus variables induces $\Phi_1 : H^*_{T\times\mathbb{C}^*,\rho_0}(W) \to H^*_{T\times\mathbb{C}^*,\rho_1}(W)$ with

$$\Phi_1(f(\lambda,\zeta)a) = f(\lambda+\zeta,\zeta)\Phi_1(a).$$

Let ι_0, ι_∞ denote the fiber inclusions. The section correspondence is defined by

$$(\widetilde{T}a, b)_\infty = \sum_{\beta,l} Q^\beta y^l \langle \iota_{0*}a, \iota_{\infty*}b\rangle^{\widehat{W},T\times\mathbb{C}^*}_{0,2,\sigma_{\min}+(\beta,l)}, \tag{40}$$

with only effective section classes included. Define $T_a = (\Phi_1^{-1}\widetilde{T})|_{\zeta=-z}$; semilinearity translates $\lambda \mapsto \lambda + z$.

Section classes. At X the normal weight is positive, so X gives $\sigma_{\min}$. A section obtained from a point of X has tautological degree zero in (39); the section from the trivial-summand fixed component B is the line $\mathcal{O}(-1)$ and has tautological degree one. Both have base class zero, so $\sigma_B - \sigma_{\min}$ is the fiber-line class of W.

The curve labels fit: on $H_2(W;\mathbb{Z})$ we have (β, l); on $H_2(\widehat{W};\mathbb{Z})$ a section class is $(\beta, 1, l)$; subtracting $\sigma_{\min} = (0,1,0)$ leaves $(\beta, 0, l)$. The fiber inclusions preserve β and tautological degree, so the two fibers share their integral identification. Splittings of a section map add integral curve classes; (38) does not collapse classes to numerical equivalence.

Proposition 5.2. *There is a shift operator*

$$T_a = A(z,\lambda,Q,y)e^{z\partial_\lambda}$$

on the small quantum cohomology module of W, whose matrix A is polynomial in z, λ at each curve coefficient in the equivariant basis. Its calibration is

$$T_a = S_W T_d S_W^{-1}, \tag{41}$$

where, writing $v = x + w_F\lambda$ for the equivariant normal roots,

$$T_d|_F = y^{j_F} \prod_x \begin{cases} x+\lambda, & F = X, \\ (x-\lambda-z)^{-1}, & F = B \end{cases} e^{z\partial_\lambda}, \quad j_X = 0,\ j_B = 1. \tag{42}$$

All operators act on the full equivariant cohomology with its super pairing.

Proof. W is smooth and projective, and (39) is projective, so its section invariants are defined by proper equivariant pushforward. The two fiber insertions are polynomial equivariant classes, and the equivariant pairing of W is perfect over $\mathbb{C}[\lambda]$, so the operator is polynomial in both equivariant parameters at each curve coefficient. Normal weights are 1 at X and -1 at B; in (38) they give the factors v and $(v-z)^{-1}$ respectively, and the section calculation gives y^{j_F}. In our fundamental-solution convention $S_W(z) = M(z)^* = M(-z)^{-1}$, so Iritani's identity $MT_a = T_d M$, with his z replaced by $-z$, is exactly (41). □

Lemma 5.3. *For each fixed base class β, the matrix coefficients of A and of $c\star$ are polynomials in y, z, λ (with no z-dependence in $c\star$). In particular they are holomorphic in y before any localization in λ.*

Proof. The dual of the bundle in (39) is the sum of the globally generated bundles $\mathrm{pr}_B^* E^*$ and $\mathrm{pr}_{\mathbb{P}^1}^* \mathcal{O}(1)$, so its tautological line bundle is globally generated and every effective section class has $l \geq 0$; subtracting $\sigma_{\min}$ does not change l.

Fix basis vectors of an input and a paired output. The projective-bundle formula on $\widehat{W}$ gives

$$\mathrm{vdim}_{\mathbb{C}} \overline{\mathcal{M}}_{0,2}(\widehat{W}, \sigma_{\min} + (\beta, l)) = \dim_{\mathbb{C}} W + 1 + (r+1)l + \int_\beta \kappa, \tag{43}$$

so virtual dimension increases by $r+1$ when l increases by one. Proper equivariant integration is polynomial in parameters. If virtual dimension exceeds total insertion degree, the result has negative cohomological degree and vanishes, so only finitely many l occur for this entry and β. The same argument with the three primary insertions defining a quantum product proves it for $c\star$. □

Lemma 5.4. $[G_a, T_a] = 0$.

Proof. By the two calibration identities it suffices to work in the descendant frame. Equivariant homogeneity of restriction gives $\lambda\partial_\lambda + \mu = \lambda\partial_\lambda + \mu_F - n_F/2$ on the F-summand, and $c|_F = c_1(TF) + \sum_x (x + w_F\lambda)$. Write $T_d|_F = a_F E_z$ with $E_z = e^{z\partial_\lambda}$. The combination $z\partial_z + \lambda\partial_\lambda$ commutes with E_z; combined with first-Hodge-degree counting it applies 1 to each $x + \lambda$ factor and -1 to each $(x-\lambda-z)^{-1}$ factor, so its commutator with a_F is $W_F a_F$, where $W_F = n_F w_F$ (with y^{j_F} held fixed). On the other hand $E_z(c|_F) = (c|_F + W_F z)E_z$ and $[c|_F/z, a_F E_z] = -W_F a_F E_z$; the two contributions cancel. Scalar-centering and cup-multiplication terms introduce no further commutators. □

5.3 The fiber spectrum of the shift

Let $Q^{>0}$ be the ideal of positive base degree. Since an effective nonzero base class has positive ample degree, $Q^{>0}$ retains exactly the maps with constant projection to B. The y-variable is retained.

Lemma 5.5. *On $H_T^*(W)$ with the fiber quantum product,*

$$T_a \bmod (z, Q^{>0}) = (p + \lambda)\star\,. \tag{44}$$

Proof. The fiber quantum product is linear over $H^*(B)$ (all genus-zero fiber evaluations have the same projection to B, so base classes pull out of pushforward), and S_W in the reduction is likewise $H^*(B)$-linear. So T_a is $H^*(B)$-linear.

Write $D_y = y\partial_y$. The divisor equation gives $S_W(zD_y - p\cup)S_W^{-1} = zD_y - p\star$, and the left operator commutes with T_d. On X, the multiplier has no y factor and $p|_X$ is independent of λ. On B, $p|_B = -\lambda$ and $[zD_y, T_d|_B] = zT_d|_B$, $[\lambda, T_d|_B] = -zT_d|_B$, so the two terms cancel there too. Reducing modulo z proves that $T_a|_{z=0}$ commutes with $p\star$.

Assign cohomological degree one to p, λ, z and degree $r+1$ to y. Polynomiality in λ forces $p \star p^k$ to equal the classical power for $k \leq r$; these powers form an $H^*(B)[\lambda]$-basis, so an $H^*(B)[\lambda]$-linear operator commuting with $p\star$ is determined by its value on 1. In this grading T_a has degree one (equivalently, (43) at $\beta = 0$ is $\dim W + 1 + (r+1)l$, and each fiber inclusion raises

insertion degree by one); so $T_a(1)|_{z=0}$ has no positive-y coefficient. At $y=0$ we have $S_W=1$, and (42) applied to 1 restricts to $p+\lambda$ on X and to zero on B; these are the restrictions of $p+\lambda$ to the fixed locus. Thus $T_a(1)|_{z=0}=p+\lambda$, and (44) follows. □

Let $I=H^{>0}(B;\mathbb{C})$; it is nilpotent, also in the presence of odd classes. Reducing the fiber quantum algebra modulo I gives

$$\mathbb{C}[\lambda,y,p]/(p^r(p+\lambda)-y). \tag{45}$$

Indeed, the classical relation reduces to $p^r(p+\lambda)=0$; in degree $r+1$ the only possible positive curve correction is a constant multiple of y. Its coefficient is the degree-one fiber line invariant:

$$\left\langle H, H^r, H^r\right\rangle^{\mathbb{P}^r}_{0,3,\text{line}}=1,$$

since two general point conditions determine a unique line and the remaining marked point meets a general hyperplane in one point; convexity makes this the virtual count as well. The coefficient has degree zero, so may be computed at $\lambda=0$.

Proposition 5.6. *For generic (y,λ), the distinct eigenvalue branches of T_a mod $(z,Q^{>0})$ are*

$$\lambda+h, \qquad h^r(h+\lambda)=y. \tag{46}$$

Each branch may have nonzero multiplicity and nilpotent part. For $\lambda\neq 0$, near $y=0$ one branch tends to 0 and a cluster of r branches tends to λ.

Proof. On $H^*(B)[\lambda,y,p]/(I)$, Lemmas 5.5 and (45) give the eigenvalues. The finite filtration by powers of I is preserved, and each graded piece of the induced operator is the same fiber multiplication tensored with I^m/I^{m+1}, so the characteristic polynomial has the same set of roots at each step. Away from the critical values of $h\mapsto h^r(h+\lambda)$, roots are distinct; at $y=0$ they are $-\lambda$ (multiplicity one) and 0 (multiplicity r). Translating by λ gives the final assertion. The argument uses only nilpotence of I, not semisimplicity of B's quantum cohomology. □

Polynomial dependence of G_a, T_a and finiteness of the spectral branches are what the next section needs. Both calibration identities kept full curve labels, so returning to $y=0$ will still compare the individual descendant theories on B and X.

6 Spectral projections and block gradings

The shift T_a has $r+1$ spectral branches at generic fiber parameter, while localization distinguishes the two fixed manifolds B and X. We construct operators on individual spectral blocks and compare their sums at $y=0$. This section is purely classical: it builds and manipulates operators (no quantized constraints yet).

6.1 Functional calculus for differential series

Fix a finite-dimensional vector space V underlying an equivariant basis of $H^*_T(W)$. On a small open U in the (y,λ)-plane, let $\mathcal{O}_U$ be holomorphic functions and set

$$\mathscr{D}_\lambda(U)=\Big\{\sum_{j=0}^{J} a_j(y,\lambda)\partial_\lambda^j : J<\infty,\ a_j\in\mathcal{O}_U\Big\}.$$

The multiplication is composition, so $\partial_\lambda a = a\partial_\lambda + \partial_\lambda(a)$; there is no y-differentiation. Let Λ_B be the base Novikov completion and $\Lambda_{B,+}$ its positive-degree ideal. We use the completed algebra

$$\mathscr{A}_U = \Big\{ \sum_{\beta,m\geq 0} Q^\beta z^m A_{\beta,m} : A_{\beta,m} \in \mathrm{End}(V) \otimes \mathscr{D}_\lambda(U) \Big\}. \tag{47}$$

Completion is by z and ample base degree. Each $A_{\beta,m}$ has finite differential order, but the order may grow with β, m. Products at a fixed coefficient are finite by Novikov finiteness. The element z is central, and $z\partial_z$ is adjoined by $[z\partial_z, Q^\beta z^m A_{\beta,m}] = mQ^\beta z^m A_{\beta,m}$.

Suppose $T \in \mathscr{A}_U$ has order-zero reduction

$$T^{(0)} = T \bmod (z, \Lambda_{B,+}) \in \mathrm{End}(V) \otimes \mathcal{O}_U.$$

After shrinking U, fix contours in the complex ξ-plane avoiding $\mathrm{Spec}\, T^{(0)}$ and enclosing specified eigenvalue groups; keep contours fixed as y, λ vary. Set $R_0(\xi) = (\xi - T^{(0)})^{-1}$ and $K = T - T^{(0)}$. The formal resolvent is

$$R_T(\xi) = (\xi - T)^{-1} = \sum_{n\geq 0} (R_0(\xi)K)^n R_0(\xi). \tag{48}$$

Each factor K raises the joint z/base filtration, so at each coefficient this is a finite matrix sum, meromorphic in ξ with poles exactly at $\mathrm{Spec}\, T^{(0)}$.

Lemma 6.1 (Coefficientwise holomorphic calculus)**.** *For T as above and a scalar holomorphic f on a neighborhood of* $\mathrm{Spec}\, T^{(0)}$ *(the neighborhood may be disconnected), the contour integral*

$$f(T) = \frac{1}{2\pi i} \int_\Gamma f(\xi) R_T(\xi)\, d\xi \tag{49}$$

is unital, multiplicative, and compatible with holomorphic composition: $g(f(T)) = (g \circ f)(T)$ when the functions are defined on the indicated neighborhoods. It gives mutually orthogonal idempotents summing to the identity (via indicator functions of spectral groups), all commuting with T.

If additionally $[T, \lambda] = zT$, then

$$[f(T), \lambda] = z(\xi f'(\xi))(T). \tag{50}$$

If a differential operator G commutes with T, acts continuously on the coefficientwise Laurent extension of (47), and commutes with the scalar spectral variable ξ, then $[G, f(T)] = 0$.

Proof. The inverse identities for (48) give the resolvent relation $R_T(\xi)R_T(\eta) = (R_T(\xi) - R_T(\eta))/(\eta - \xi)$ (both spectral variables central). Nested-contour integration and scalar Cauchy give multiplicativity. Unitality follows directly from (48). For the composition rule, apply the calculus to $(\eta - f(\xi))^{-1}$ at η outside $\mathrm{Spec}\, f(T^{(0)})$: the product with $\eta - f(T)$ is the identity, so it is the resolvent of $f(T)$; integrating against $g(\eta)$ gives $g(f(T)) = (g \circ f)(T)$.

For (50), use the inverse rule: $[R_T(\xi), \lambda] = R_T(\xi)[T, \lambda]R_T(\xi) = z(\xi R_T(\xi)^2 - R_T(\xi)) = -z\partial_\xi(\xi R_T(\xi))$. Integration by parts gives (50). Finally, $[G, T] = [G, \xi] = 0$ implies $[G, R_T(\xi)] = 0$ by the inverse identity; integration gives $[G, f(T)] = 0$. Contours held fixed locally allow coefficientwise differentiation under the integral. □

We will need a larger algebra for frame changes: $\mathcal{O}$ a ring of parameter germs stable under ∂_λ, $\mathcal{M}$ the base Novikov monoid or the full monoid $\{(\beta, l) : l \geq 0\}$:

$$\mathscr{A}_{\mathcal{M}}^{\mathrm{Laur}}(\mathcal{O}) = \Big\{ \sum_{d \in \mathcal{M}} q^d \sum_{m \geq m(d)} z^m A_{d,m} : A_{d,m} \in \mathrm{End}(V) \otimes \mathscr{D}_\lambda(\mathcal{O}),\ m(d) \in \mathbb{Z} \Big\}. \tag{51}$$

Here q^d means Q^β or $Q^\beta y^l$. Multiplication is well-defined: each fixed d has finitely many decompositions, and the two Laurent lower bounds leave finitely many z-summands at a fixed power.

Lemma 6.2 (Change of frame)**.** *Let T, T' have coefficientwise resolvents on a common contour as in Lemma 6.1, both in (51). Suppose M, M^{-1} lie in the same algebra, do not depend on ξ, and $T' = MTM^{-1}$. Then*

$$R_{T'}(\xi) = M R_T(\xi) M^{-1}, \qquad f(T') = M f(T) M^{-1}$$

for every f from the contour calculus.

Proof. $M R_T(\xi) M^{-1}$ is a two-sided inverse of $\xi - T'$; integrate coefficientwise and use the Laurent lower bounds to keep products finite at each coefficient. □

Two forms of M will occur. First, an identity-modulo-positive- Novikov matrix whose coefficients are Laurent expansions at $z = 0$ of rational functions of z (its Novikov inverse shares this property). Second, $\exp(N/z) M_+(z)$ with N a nilpotent cup-product and $M_+(z), M_+^{-1}$ regular at $z = 0$; both forms satisfy the Laurent requirement.

6.2 Block operators without λ-differentiation

We apply the calculus to $T = T_a$. By Lemma 5.3, T_a's coefficients are polynomial in y, λ, z at each fixed base degree (before the shift $\exp(z\partial_\lambda)$ is expanded), so $T_a \in \mathscr{A}_U$. By Proposition 5.6, the spectrum of the reduction $T_a^{(0)}$ is

$$a_i = \lambda + h_i, \qquad h_i^r(h_i + \lambda) = y. \tag{52}$$

Take $\lambda \neq 0$, $y \neq 0$, exclude roots coinciding, and shrink to a small open U. Choose logarithms near each a_i (a whole generalized eigenspace; no diagonalizability assumed). Define

$$P_i = \frac{1}{2\pi i} \int_{\gamma_i} R_{T_a}(\xi)\, d\xi, \quad \mathscr{L}_i = \frac{1}{2\pi i} \int_{\gamma_i} \log \xi\, R_{T_a}(\xi)\, d\xi, \tag{53}$$

where γ_i encloses a_i with its multiplicity. For a group of eigenvalues use the sum of contours.

Proposition 6.3 (Removal of coefficient differentiation)**.** *The projectors and logarithms satisfy*

$$[P_i, \lambda] = 0, \qquad [\mathscr{L}_i, \lambda] = z P_i, \qquad [G_a, P_i] = [G_a, \mathscr{L}_i] = 0. \tag{54}$$

In particular P_i and $M_i = \mathscr{L}_i - z P_i \partial_\lambda$ are matrix series without coefficient differentiation. The operator

$$D_i = P_i G_a - \frac{\lambda}{z} \mathscr{L}_i \tag{55}$$

differentiates only in z, and $P_i D_i = D_i P_i = D_i$.

Proof. $[T_a, \lambda] = zT_a$. Apply (50) to the locally-constant indicator and its supported logarithm. $[G_a, T_a] = 0$ and $[G_a, \xi] = 0$ give $[G_a, P_i] = [G_a, \mathscr{L}_i] = 0$.

At each coefficient of P_i and $\mathscr{L}_i - zP_i\partial_\lambda$, the identity $[D, \lambda] = 0$ for finite-order D forces D to have order zero (the leading coefficient of $[a_J\partial_\lambda^J, \lambda]$ is $Ja_J\partial_\lambda^{J-1}$). Writing out G_a gives

$$D_i = P_i z\partial_z + P_i(\tfrac{1}{2} + \mu) + z^{-1}(P_i(c\star) - \lambda M_i). \tag{56}$$

No λ-derivatives remain, and neither G_a nor the functional calculus introduced derivatives in y or Q. Support $P_iD_i = D_iP_i = D_i$ follows from $[G_a, P_i] = 0$, $P_i\mathscr{L}_i = \mathscr{L}_iP_i = \mathscr{L}_i$, and $[P_i, \lambda] = 0$. □

Proposition 6.4 (Modes and coefficient bounds)**.** *Define*

$$A_{-1,i} = z^{-1}P_i, \qquad A_{k,i} = z^{-1}(zD_i)^{k+1} \quad (k \geq 0). \tag{57}$$

Each $A_{k,i}$ is a series in powers z^m, $m \geq -1$, with coefficients that are matrix differential operators in $z\partial_z$ of order at most $k+1$ and contain no derivatives in y, λ, Q. Each fixed base and loop coefficient extends holomorphically along paths in the generic locus of (52), with the spectral labels and logarithms continued along the path.

For any scalar b independent of z, replacing D_i by $D_i + bz^{-1}P_i$ gives

$$A_{k,i} \longmapsto \sum_{j=0}^{k+1} \binom{k+1}{j} b^{k+1-j} A_{j-1,i}. \tag{58}$$

For distinct branches $i \neq j$: $(zD_i)(zD_j) = 0$. Consequently, if $P = \sum_i P_i$ and $D = \sum_i D_i$ over a group of branches, the modes formed from (P, D) are the sum of the branches' modes.

Proof. Equation (56) shows zD_i has nonnegative z-powers and Euler order at most one; products of $z\partial_z$ preserve z-powers. Holomorphic continuation at each fixed coefficient uses that the resolvent construction involves finitely many inversions, parameter derivatives, and contour residues; its only spectral denominators are the a_i, which continue along any path on which they are distinct and nonzero. This is a per-coefficient statement; no global convergence asserted.

Since D_i commutes with b and P_i is a two-sided identity for D_i, the ordinary binomial formula applies in this supported algebra, giving (58). Finally, $(zD_i)(zD_j) = zD_iP_izP_jD_j = 0$ for $i \neq j$, and quantization is linear in the modes. □

Changing a logarithm by $2\pi in$ changes D_i by $-2\pi in\lambda z^{-1}P_i$; the modes for one logarithm and any other span the same collection. Infinitesimal symplecticity will follow from the fixed-manifold comparison below; it is not needed for the constructions here.

6.3 Fixed-manifold blocks at $y = 0$

Fix a germ of $\lambda \neq 0$. At $y = 0$ one eigenvalue $a_B = \lambda + h_B$ tends to zero and the remaining r tend to λ. Call the small branch the B *branch* and the r-element group the X *cluster*. Choose disjoint contours about 0 and λ enclosing these groups for y small. Both projectors extend through $y = 0$; the logarithm applies through $y = 0$ on the X contour (which avoids 0). Denote the resulting operators $P_B, P_X, \mathscr{L}_X$.

Switch the fixed-manifold coordinates (restriction to $B \sqcup X$); the change is invertible at $\lambda \neq 0$ and preserves the completions. Let Π_B, Π_X be the constant coordinate projectors and put

$$K_F(z,\lambda) = \prod_x \begin{cases} x+\lambda, & F = X, \\ (x-\lambda-z)^{-1}, & F = B, \end{cases} \qquad E_z = e^{z\partial_\lambda},$$

where x ranges over the ordinary Chern roots of N_F. Then (42) reads $T_d|_F = y^{j_F} K_F E_z$.

Proposition 6.5 (Comparison at zero fiber degree)**.** *The projectors P_B, P_X and the cluster logarithm $\mathscr{L}_X$ have holomorphic coefficient germs through $y = 0$. Their Taylor expansions at $y = 0$ satisfy*

$$P_F = S_W \Pi_F S_W^{-1} \ (F = B, X), \qquad \mathscr{L}_X = S_W \mathscr{L}_{d,X} S_W^{-1}, \tag{59}$$

where $\mathscr{L}_{d,X}$ is the log of $K_X E_z$ on the X summand, extended by zero on B.

$T_a P_B$ is divisible by y coefficientwise in z and base degree. Set $\widetilde{T}_B = y^{-1} T_a P_B$. It has holomorphic coefficient germs through $y = 0$. Its nonzero block at $y = z = Q^{>0} = 0$ has eigenvalue $(-\lambda)^{-r}$ (possible nilpotent part); let $\mathscr{K}_B$ be its supported logarithm. Then on a punctured sector in y one can choose the B-branch logarithm as

$$\mathscr{L}_B = (\log y) P_B + \mathscr{K}_B. \tag{60}$$

The coefficients of $\mathscr{K}_B$ are holomorphic through $y = 0$. Its Taylor expansion, and hence (60), obeys the same conjugation by S_W as in (59), with the fixed-block operators in the descendant frame.

Proof. The reduction $T_a^{(0)}$ is a holomorphic matrix through $y = 0$; its spectrum there consists of 0 and λ (including multiplicities and nilpotent parts). The Neumann construction with the two fixed contours gives the asserted holomorphic extensions.

Taylor expanding coefficient germs in y is a homomorphism commuting with ∂_λ; it sends the ancestor resolvent (nonnegative loop powers) to a Laurent series in the full Novikov labels and preserves the inverse identities. Use (51) with $d = (\beta, l)$, $q^d = Q^\beta y^l$, and coefficients holomorphic in the chosen λ-germ. Rationality in Proposition 4.1 puts the expansion of S_W at $z = 0$ in that algebra; its zero-curve coefficient is the identity, so its inverse also sits in it. The shift operators expanded via (42) also lie in it, so their resolvents on the two contours lie in it. The descendant resolvent uses the joint filtration by z and full Novikov degree: its initial B block is zero and its initial X block is $\lambda + p$, whose nilpotent part is the ordinary class $p|_X$.

(41) and Lemma 6.2 now identify the actual resolvents in the two frames. In the descendant frame the contour about zero selects the whole B summand, and the contour about λ selects the whole X summand (within each summand the contour function is identically 1 or 0, so Lemma 6.1 gives identity or zero also for the formal perturbation). Integration gives (59), including logarithms on X.

On the B summand $T_d \Pi_B = y K_B E_z \Pi_B$; y is central. So $T_a P_B = y S_W K_B E_z \Pi_B S_W^{-1}$ in the Laurent algebra completed in full Novikov degree. Neither S_W nor its inverse has negative y-powers, so the constant Taylor coefficient in y of every base and loop coefficient on the left vanishes. Those coefficients are already holomorphic through $y = 0$, so division gives $\widetilde{T}_B = S_W K_B E_z \Pi_B S_W^{-1}$ (after Taylor expansion).

At $y = z = Q^{>0} = 0$, the calibration is the identity and the nonzero block of the previous identity is $\prod_x (x - \lambda)^{-1}$. Ordinary positive-degree classes are nilpotent, so the sole spectral

value is $(-\lambda)^{-r} \neq 0$. Choose a contour about this value avoiding zero and a logarithm there; Lemma 6.1 defines $\mathscr{K}_B$ holomorphically through $y = 0$ with the projector P_B as its two-sided identity (by Proposition 6.5 and Lemma 6.2).

Finally, restrict to a punctured sector and choose $\log y$. On the P_B-supported algebra $T_a = y\widetilde{T}_B$. Rescaling the spectral contour by the central scalar y and choosing $\log(y\xi) = \log y + \log \xi$ shows that its logarithm is exactly (60). Conjugation follows because multiplication by $\log y$ commutes with everything. □

Near $y = 0$ the B-branch modes are finite polynomials in $\log y$ with holomorphic coefficients: write $D_B = D_B^{\mathrm{reg}} - \lambda(\log y)z^{-1}P_B$ with $D_B^{\mathrm{reg}} = P_B G_a - (\lambda/z)\mathscr{K}_B$ holomorphic, then apply (58) with $b = -\lambda \log y$. For $A_{k,B}$ the logarithmic degree is at most $k+1$. The X-cluster modes are holomorphic through zero. None of this requires convergence in z or an analytic meaning for the Novikov series.

7 The ordinary theory on a fixed component

The operators of Section 6 were built from the equivariant theory of W. This section identifies what their constraints mean near $y = 0$: on a fixed component, Quantum Riemann–Roch removes the inverse-Euler twist, and the logarithm of the shift becomes a translation generator in λ. Its derivative term cancels the derivative term of the equivariant grading, leaving the ordinary grading plus a scalar multiple of z^{-1}.

Proposition 7.1. *Fix a germ at $\lambda \neq 0$. Near $Q = y = 0$, use the full Novikov completion with a formal $\log y$ adjoined, allowing polynomial dependence on $\log y$ in each coefficient. Let $F = B$ denote the branch near the eigenvalue zero, or $F = X$ the whole cluster near the eigenvalue λ. Use the projectors and logarithms of Proposition 6.5, and let $A_{k,F}$ be their modes. Then, over this coefficient ring,*

$$\mathcal{A}_W \text{ satisfies every } A_{k,F} \text{ projectively} \iff Z_F \text{ satisfies every } \ell_{k,F} \text{ projectively}, \quad k \geq -1.$$

The operators on the left are infinitesimal symplectic for the equivariant pairing of W. On the right, Z_F is the ordinary descendant potential with its full cohomology and actual integral curve labels.

The proof has two parts. First we show that the fixed-block operators, after a change of paired space, become the ordinary Virasoro modes plus a scalar shift. Then we show that this change of space is realized by the quantum Riemann–Roch multiplier and, after separating the rank factor, that the resulting quantized equations are those of the actual Euler twist.

7.1 Removing the normal bundle from the grading

Fix F and abbreviate $n = n_F$, $w = w_F$, $j = j_F$. Put

$$a = w\lambda, \qquad c_N = c_1(N_F)_{\mathrm{ord}}, \qquad W_F = nw.$$

After restriction to the descendant space of F, the grading and shift from Section 5 are

$$G_F = z\partial_z + \lambda\partial_\lambda + \tfrac{1}{2} + \mu_F - \tfrac{n}{2} + \frac{R_F + c_N + nw\lambda}{z}, \tag{61}$$

$$T_F = y^j \prod_x \begin{cases} x + \lambda, & w = 1, \\ (x - \lambda - z)^{-1}, & w = -1 \end{cases} e^{z\partial_\lambda}. \tag{62}$$

Here x runs over the ordinary Chern roots of N_F; symmetric expressions are read via the splitting principle, and all x-expansions are finite on $H^*(F)$.

Choose a germ of $\log a$ at a nonzero λ. Define a multiplier Δ_F by

$$\log \Delta_F = \sum_x \Big\{ \tfrac{1}{2} \log(a+x) + \frac{1}{z} \int_0^x -\log(a+t)\, dt + \sum_{m \geq 1} \frac{B_{2m}}{(2m)!} z^{2m-1} \partial_x^{2m-1} \big(-\log(a+x)\big) \Big\}, \quad (63)$$

with B_{2m} the Bernoulli numbers. The z^{-1}-coefficient has positive ordinary cohomological degree, hence is a nilpotent multiplication; both Δ_F and Δ_F^{-1} have a finite z-lower bound, and positive tails are formal power series at $z = 0$.

The multiplier maps the ordinary paired loop space of F to the space with pairing η_F^t. Indeed, all terms of (63) except the half-logarithm are odd in z, so

$$\Delta_F(-z)\Delta_F(z) = \prod_x (a+x) = e_T(N_F),$$

and the inverse Euler factor in η_F^t cancels this product. This fixes the direction of the square-root factor.

For the scalar part of the shift, put

$$L(v) = v - v \log v, \qquad \Theta_F(\lambda) = nL(w\lambda). \quad (64)$$

Lemma 7.2. *The following hold in the Laurent calculus of Lemma 6.2:*

$$\Delta_F^{-1} G_F \Delta_F = \ell_{0,F} + \lambda \partial_\lambda + \frac{nw\lambda}{z}, \quad (65)$$

$$\Delta_F^{-1} T_F \Delta_F = y^j \exp\Big(\frac{\Theta_F(\lambda) - \Theta_F(\lambda + z)}{z} \Big) e^{z\partial_\lambda}. \quad (66)$$

Under the same change of space, the logarithm prescribed near $y = 0$ becomes

$$z\partial_\lambda + j \log y - \Theta_F'(\lambda) = z\partial_\lambda + j \log y + nw \log(w\lambda), \quad (67)$$

up to a scalar constant set by the logarithm branch.

Proof. The cohomological commutator with μ_F differentiates functions of a root by $x\partial_x$ because x has Hodge type $(1,1)$. Apply $z\partial_z + \lambda\partial_\lambda + x\partial_x$ to a summand of (63). The half-logarithm contributes $1/2$. For $I(a,x) = \int_0^x -\log(a+t)dt$ we have $(\lambda\partial_\lambda + x\partial_x)I = I - x$; its contribution is $-x/z$. Each Bernoulli term has total degree zero. Summing over roots gives $n/2 - c_N/z$, canceling the two corresponding terms in (61), giving (65).

For the shift, add $L(a)/z$ to the summand indexed by x in (63). The sum depends only on $v = a + x$ and equals

$$\Phi(v) = \Big((z\partial_v)^{-1} - \tfrac{1}{2} + \sum_{m \geq 1} \frac{B_{2m}}{(2m)!} (z\partial_v)^{2m-1} \Big)(-\log v),$$

where the antiderivative is $L(v)/z$. The Bernoulli generating series gives $\Phi(v+z) - \Phi(v) = -\log v$. When $w = 1$, this difference cancels the factor v in (62); when $w = -1$, use $\Phi(v - z) - \Phi(v) = \log(v - z)$ to cancel the factor $(v - z)^{-1}$. The added terms $L(a)/z$ account for the exponential in (66).

For the logarithm (as opposed to the exponential form), set $H = z\partial_\lambda - \Theta'_F(\lambda)$. Its evolution is

$$e^{tH} = \exp\Big(\frac{\Theta_F(\lambda) - \Theta_F(\lambda + tz)}{z}\Big) e^{tz\partial_\lambda}; \tag{68}$$

differentiating the right side in t gives H times that side, and the value at $t = 0$ is the identity. The functional calculus of Lemma 6.1 agrees (coefficientwise differentiation of the contour integral gives the same evolution). The z-constant term is $-\Theta'_F(\lambda)$; locally choose a logarithm that inverts the exponential at that scalar; the composition rule in Lemma 6.1 then identifies $\log(e^H)$ with H. Conjugating by Δ_F is allowed by Lemma 6.2. For $F = B$, apply this to the divided shift $y^{-1}T_F$ and restore $\log y$; for $F = X$, apply it directly to T_F. These are exactly the logarithm prescriptions of Proposition 6.5, giving (67) up to the logarithm-branch scalar. □

Subtracting λ/z times (67) from (65) removes $\lambda\partial_\lambda$. Thus the operator D_F, after the descendant change of space and then Δ_F, becomes

$$\ell_{0,F} + \frac{b_F}{z}, \qquad b_F = \lambda(nw - j\log y - nw\log(w\lambda)), \tag{69}$$

up to a scalar multiple of λ/z from a logarithm choice. In particular it differentiates neither λ, y, nor the Novikov variables.

For a scalar b independent of z, write

$$\ell^{(b)}_{-1,F} = z^{-1}, \qquad \ell^{(b)}_{k,F} = z^{-1}(z\ell_{0,F} + b)^{k+1} \quad (k \geq 0).$$

The binomial identity

$$\ell^{(b)}_{k,F} = \sum_{m=-1}^{k} \binom{k+1}{m+1} b^{k-m}\ell_{m,F} \tag{70}$$

is triangular with diagonal entries one. Consequently these operators are infinitesimal symplectic, and their simultaneous projective equations are equivalent to those of the ordinary modes. This completes the classical identification. Now we lift it to the potentials.

7.2 Quantizing the comparison

The Δ_F above contains $-\log(w\lambda)c_N/z$, a classical multiplication that we cannot directly exponentiate in λ^{-1}. We therefore first use the normalized class

$$c_0(N_F) = \prod_x (1 + x/a)^{-1}, \tag{71}$$

with $Z_{F,\mathrm{ntw}}$ and η_F^{ntw} the corresponding descendant potential and pairing. Define $\Delta_{F,0}$ by (63) with $-\log(a + x)$ replaced by $-\log(1 + x/a)$ and $\frac{1}{2}\log(a + x)$ by $\frac{1}{2}\log(1 + x/a)$. It maps the ordinary paired space to $(\mathcal{H}_F, \eta_F^{\mathrm{ntw}})$ and is the identity modulo λ^{-1}.

Lemma 7.3 (Quantum Riemann–Roch in the normalized twist)**.** *With quantization and dilaton translations as in Section 2,*

$$Z_{F,\mathrm{ntw}} = (\textit{invertible scalar})\, U_{\Delta_{F,0}} Z_F. \tag{72}$$

This is an invertible identity formal in λ^{-1} and coefficientwise in the full Novikov variables and $\hbar$.

Proof. Quantum Riemann–Roch of Coates–Givental says that for a multiplicative class whose logarithm on a line with first Chern class x is $s(x)$, its multiplier (in the fixed ordinary pairing) has exponent

$$\sum_x \Big\{ \frac{1}{z} \int_0^x s(t)dt + \sum_{m\geq 1} \frac{B_{2m}}{(2m)!} z^{2m-1} s^{(2m-1)}(x) \Big\},$$

and it identifies the twisted and ordinary pairings by multiplication by $\sqrt{c_0(N_F)}$. Expressing the conclusion as a map *to* the actual twisted space adds $-s(x)/2$ to the exponent. Taking $s(x) = -\log(1+x/a)$ gives exactly $\Delta_{F,0}$.

The affine-coordinate bookkeeping matches: in the ordinary Fock space the twisted potential uses $\widetilde{q} = \sqrt{c_0(N_F)}(t - z1_F)$, and multiplication by $c_0(N_F)^{-1/2}$ returns the twisted Darboux coordinate $q_{\mathrm{ntw}} = t - z1_F$. Thus $U_{\Delta_{F,0}}$ has the dilaton translations of (72). The theorem applies to $R\pi_* \operatorname{ev}^* N_F$ on the ordinary stable-map spaces of the smooth projective F, is invertible and formal in λ^{-1}, and uses the categorical inverse for nodal contractions. So it holds in our Koszul conventions, for arbitrary Hodge types and parity. □

The two classical multipliers are related simply:

$$\Delta_F = a^{n/2} \exp(-\log(a) c_N/z)\, \Delta_{F,0}. \tag{73}$$

If C is an ordinary divisor class, then $[\ell_{0,F}, C/z] = [z\partial_z, C/z] + [\mu_F, C/z] = 0$ and $[z, C/z] = 0$ (the two contributions being $-C/z$ and $+C/z$, since R_F commutes with C). So C/z commutes with all ordinary modes and with their scalar translates (70). After the coefficient derivatives are removed in (69), the scalar $a^{n/2}$ also cancels in conjugation. Consequently Δ_F and $\Delta_{F,0}$ produce the same conjugated mode matrices. By Lemma 7.3 and Lemma 2.2, the equations for those matrices on $Z_{F,\mathrm{ntw}}$ are equivalent to the ordinary projective equations on Z_F.

It remains to restore the rank factor omitted in (71). This changes the paired space as well as the invariants, and both must be handled together.

Lemma 7.4 (Rank scaling)**.** *Put $m_F = a^{-n}$. The actual inverse-Euler twist is obtained from the normalized twist by the substitutions*

$$\hbar \longmapsto \hbar/m_F, \qquad Q^\beta y^l \longmapsto Q^\beta y^l a^{-\int_d c_N} \tag{74}$$

(where d is the corresponding actual class on F). Its pairing is $\eta_F^t = m_F \eta_F^{\mathrm{ntw}}$. Under these substitutions the quadratic quantizations of $\Delta_{F,0}\ell_{k,F}\Delta_{F,0}^{-1}$ (for every $k \geq -1$) in the two paired spaces agree.

Proof. On $\overline{\mathcal{M}}_{g,m}(F,d)$, Riemann–Roch gives $\operatorname{rank} R\pi_* \operatorname{ev}^* N_F = n(1-g) + \int_d c_N$. The ratio of the inverse-Euler class to the normalized class is therefore $a^{-n(1-g)-\int_d c_N} = m_F^{1-g} a^{-\int_d c_N}$, exactly the change in the coefficient of $\hbar^{g-1} Q^\beta y^l$ under (74).

The qq part of quantization uses $\eta/\hbar$; the pp part uses $\hbar\eta^{-1}$. In the actual twist these become $m_F\eta^{\mathrm{ntw}}/\hbar$ and $(\hbar/m_F)(\eta^{\mathrm{ntw}})^{-1}$, exactly the normalized counterparts at $\hbar/m_F$. The mixed part is unchanged. Since $\Delta_{F,0}\ell_{k,F}\Delta_{F,0}^{-1}$ has no Novikov derivatives or Novikov dependence, the curve substitution commutes with its action. Restore the scalar b_F afterwards by (70). Each curve monomial is multiplied by a nonzero scalar, so the substitution is injective and reversible without identifying curve classes. □

Rationality in λ. The above used QRR only as a formal identity in λ^{-1}. To pass to germs at nonzero λ, first remove b_F by (70). Each coefficient of the conjugated matrices $\Delta_{F,0}\ell_{k,F}\Delta_{F,0}^{-1}$ is then rational in λ; the matrices have finite z-lower bounds. The coefficients of $Z_{F,\mathrm{tw}}$ are also rational in λ (the torus acts trivially on F, the inverse Euler class expands only up to finite cohomological dimension, and descendant indices are bounded at fixed genus, class and number of marks). Then each nonconstant coefficient of $Z_{F,\mathrm{tw}}^{-1}\operatorname{op}_{\eta_F^t}(\Delta_{F,0}\ell_{k,F}\Delta_{F,0}^{-1})Z_{F,\mathrm{tw}}$ is a finite sum of rational functions of λ; vanishing of its Laurent expansion at infinity is equivalent to vanishing as a germ. At each genus and curve coefficient, (74) only shifts finitely many λ-powers, so the argument runs in both directions. We then restore b_F and its logarithms via (70). No analytic value of the quantized infinite multiplier is taken.

7.3 Proof of Proposition 7.1

Let F' be the other fixed component. Up to invertible scalars, the potential chain is

$$Z_F \xrightarrow{U_{\Delta_{F,0}}} Z_{F,\mathrm{ntw}} \xrightarrow{(74)} Z_{F,\mathrm{tw}},$$
$$Z_{F,\mathrm{tw}} \xrightarrow{U_{S_f}} \mathcal{A}_{F,\mathrm{tw}}(u_F),$$
$$\mathcal{A}_{F,\mathrm{tw}}(u_F)\,\mathcal{A}_{F',\mathrm{tw}}(u_{F'}) \xrightarrow{U_R} \mathcal{A}_W.$$

The arrow (74) changes parameters and rescales the pairing per Lemma 7.4. The final row first adjoins the other fixed potential and then applies the upper symplectic transformation.

(70), Lemma 7.3 and Lemma 7.4 identify projective satisfaction of all ordinary modes on Z_F with projective satisfaction of the corresponding descendant modes on $Z_{F,\mathrm{tw}}$. The rationality argument above gives the equivalence over the same germs in λ used by the spectral construction.

Next, pass from descendants on F to ancestors at u_F: the ancestor–descendant identity (16) and Lemma 2.2 conjugate the mode matrices by $S_f = S_{F,\mathrm{tw}}(u_F, z)$ and transport their projective equations to $\mathcal{A}_{F,\mathrm{tw}}(u_F)$. On the product of the two fixed potentials, an operator supported on F acts only on that factor, so its projective equation is equivalent to the equation on the single factor.

Finally, apply $R(z)$ of Proposition 4.1. The identities

$$S_W = R(z)\bigoplus_F S_f, \qquad \mathcal{A}_W = (\text{scalar})\,U_R \prod_F \mathcal{A}_{F,\mathrm{tw}}(u_F)$$

show, by Proposition 6.5, that the resulting classical modes are exactly $A_{k,F}$ and their quantum equations are the projective equations on $\mathcal{A}_W$. Each change is invertible, so the implication holds both ways.

All quantized changes are defined in the Novikov completion: u_F, $S_f - I$, and $R - I$ have positive Novikov filtration; S_f has a finite negative tail at each curve coefficient; R is upper. Conjugation and quantization preserve the lower loop bounds and finite support required in Section 2. The comparison uses the upper quantization from localization plus the fixed-theory ancestor changes; it does not require a new expansion of S_W. □

8 Continuation and the ordinary constraints

The fixed-component comparison converts the hypothesis on B into equations on one spectral branch of the ancestor theory of W. We continue them to every branch. Finiteness comes from

three sources: each coefficient test is a finite sum (via ancestor bounds); continuation of the branch is holomorphic in each coefficient; and the branch cover is connected outside its critical values. Then adding the r branches of X and using the same comparison gives the ordinary constraints on X.

8.1 Polynomial coefficients and ancestor bounds

Write $\log \mathcal{A}_W = \sum_{g\geq 0} \hbar^{g-1}\mathcal{F}_g$ for the connected ancestor potential at background zero, taken in the polynomial equivariant basis of Section 3 (no localization of invariants themselves).

Lemma 8.1 (Polynomial coefficients and ancestor bounds)**.** *Fix a genus g, an actual base class β, and insertions $b_i\bar{\psi}_i^{a_i}$, $1 \leq i \leq n$. Their connected invariant in W, summed over fiber degree with weight y^l, is polynomial in y and λ. It vanishes unless the distinguished curve is stable and*

$$\sum_i a_i \leq 3g - 3 + n. \tag{75}$$

The bound is independent of β, l, λ.

Proof. Curve-unstable ancestor terms are omitted by definition. For stable data the cotangent-class product is pulled back from $\overline{\mathcal{M}}_{g,n}$, of dimension $3g-3+n$, so (75) holds before integration, without any equivariant degree input.

The virtual dimension of $\overline{\mathcal{M}}_{g,n}(W,(\beta,l))$ is

$$(1-g)(\dim_{\mathbb{C}} W - 3) + n + \int_\beta \kappa + (r+1)l, \qquad \kappa = c_1(TB) + c_1(E),$$

and insertion degrees are fixed. Proper equivariant pushforward lands in $\mathbb{C}[\lambda]$ (no negative cohomological degree), so the integral vanishes once the virtual dimension exceeds total insertion degree. Since $l \geq 0$, only finitely many l contribute at fixed β, g, n. Each remaining integral is polynomial in λ. □

8.2 Equations as germs and finite coefficient tests

Modes on the B branch are infinitesimal symplectic at $y = 0$ by Proposition 7.1; their entries are finite $\log y$-polynomials with coefficients holomorphic at zero (Proposition 6.5). Taylor expansion is injective on this ring, so the symplectic identities hold as germs on a punctured sector; they continue entrywise along any path on which the modes themselves continue.

For a continued branch and logarithm, set

$$\mathcal{E}_{k,i} = \mathcal{A}_W^{-1}\,\mathrm{op}(A_{k,i})\mathcal{A}_W, \tag{76}$$

meaning the operator acts on $\mathcal{A}_W$ then multiplies by $\mathcal{A}_W^{-1}$. For the entire X cluster, $\mathcal{E}_{k,X}$ uses $A_{k,X} = \sum_{i\in X} A_{k,i}$ with a single logarithm. Projective satisfaction means every positive- degree (insertion) coefficient of (76) vanishes.

Lemma 8.2 (Finite coefficient tests)**.** *Fix $k \geq -1$, $\hbar^{G-1}$, base class β, and an insertion monomial of total degree N in (76). Its coefficient is a finite sum of holomorphic germs on the spectral covering of generic (y,λ), with logarithms as chosen, and continues along every path in that locus. Near $y = 0$ the coefficient of $\mathcal{E}_{k,B}$ is a finite $\log y$-polynomial with coefficients holomorphic through zero; the coefficient of $\mathcal{E}_{k,X}$ is holomorphic through zero. Its Taylor expansion is the corresponding formal Novikov coefficient test.*

Proof. Let $F = \log \mathcal{A}_W$ in positive loop coordinates q. A normally ordered quadratic operator has schematic form

$$\frac{1}{2\hbar} C(q,q) + \sum_{\alpha,\gamma} B_{\alpha\gamma} q^\gamma \partial_\alpha + \frac{\hbar}{2} \sum_{\alpha,\gamma} D_{\alpha\gamma} \partial_\alpha \partial_\gamma,$$

with graded-symmetric tensors. Dividing its action on e^F by e^F replaces ∂_α by $\partial_\alpha F$ and $\partial_\alpha \partial_\gamma$ by $\partial_\alpha \partial_\gamma F + (\partial_\alpha F)(\partial_\gamma F)$.

At $\hbar^{G-1}$: the first-derivative term involves $\mathcal{F}_G$; the second-derivative term involves $\mathcal{F}_{G-1}$ and products $\mathcal{F}_g \mathcal{F}_h$ with $g + h = G$; the multiplication term occurs only at $\hbar^{-1}$. For a fixed output monomial, the connected potential has at most $N + 2$ distinguished marks, and Lemma 8.1 bounds every differentiated descendant index uniformly in l.

By Lemma 6.4, $A_{k,i}$ has loop powers at least -1, finite $z\partial_z$ order; at a fixed test monomial, the pp part uses only bounded indices; the pq part uses a bounded differentiated index plus a multiplied index in the output monomial or the dilaton; the qq part bounds index pairs by the order in $z\partial_z$. So each part uses finitely many loop coefficients of the operator, and the z derivatives only multiply by polynomials in the mode indices.

At fixed β, Novikov finiteness leaves finitely many decompositions into base classes of the operator and the $\mathcal{F}_g$'s. The $\mathcal{F}_g$'s are polynomials in y (Lemma 8.1); operator entries are holomorphic germs continuing with the branches (Lemma 6.4); the pairing and its inverse are fixed rational matrices in λ (we work at $\lambda \neq 0$). Finiteness, holomorphy and continuation follow.

Finally, Proposition 6.5 gives: X-cluster coefficients holomorphic through zero; B-branch coefficients polynomial in $\log y$ with coefficients holomorphic through zero. All the operations in the coefficient test are finite, so Taylor expansion commutes with the test and yields the formal identity of Proposition 7.1. □

A finite sum $\sum_{j=0}^{J} f_j(y)(\log y)^j$, f_j holomorphic at zero, whose formal series in y and $\log y$ vanishes is equal to zero on a punctured sector (each Taylor coefficient of every f_j vanishes). This is the bridge from formal to analytic (coefficient by coefficient).

8.3 Transport between fixed components

Fix $\lambda \neq 0$. The branch cover is

$$y = h^r(h + \lambda). \tag{77}$$

Its critical values are 0 and $(-r)^r \lambda^{r+1}/(r+1)^{r+1}$. Removing those gives an unramified covering of degree $r + 1$.

Lemma 8.3. *The monodromy of (77) acts transitively on its $r + 1$ branches.*

Proof. The inverse image of the complement of the critical values is the h-plane with finitely many points removed; it is connected and path connected. Given two points over a regular value, join them by a path in this inverse image; its projection is a loop at that value whose lift connects the two points. That is transitivity. □

Proposition 8.4. *If the ordinary descendant potential of B satisfies its ordinary modes projectively, then the same holds for X.*

Proof. By Proposition 7.1, the hypothesis gives all projective equations for $A_{k,B}$ on $\mathcal{A}_W$ near $y = 0$. Lemma 8.2 turns their formal coefficient identities into identities of germs on a punctured

sector; the B-branch infinitesimal symplecticity also holds there as germs of the fixed comparison, and is entrywise linear so continues with the operators.

Choose a regular value in this sector. Every coefficient of every projective equation continues along any loop based there. The ancestor coefficients themselves return unchanged (polynomial in y at each fixed base class). The only change is in the continued spectral branch and logarithm of its modes. By Lemma 8.3, the continued B branch can be any of the $r+1$ branches. Hence projective equations and symplecticity hold on each branch.

Continuing a logarithm may add $2\pi i\mathbb{Z}$ times P-supported λz^{-1}; by Lemma 6.4 this is an invertible triangular binomial change. The simultaneous equations therefore hold for any chosen local logarithm on each branch.

Return to a punctured neighborhood of $y = 0$ and choose one logarithm on the whole X cluster. Its projectors sum to P_X, its logarithm to $\mathscr{L}_X$; in particular $D_X = \sum_{i\in X} D_i$. Since $(zD_i)(zD_j) = 0$ for $i \neq j$, the sum's modes are the sums of the branches' modes, and quantization is linear in the modes, so the constant from projective satisfaction is insertion-independent.

Proposition 6.5 extends the cluster operators holomorphically through $y = 0$. Taylor coefficients of the equations are legitimate by Lemma 8.2. Then Proposition 7.1 applied in the reverse direction with $F = X$ gives the result for Z_X with the full curve labels. □

8.4 The prescribed scalar normalization

Projective equations carry insertion-independent scalars. For the ordinary Virasoro operators, two commutators remove them, and the characteristic constant C_Y is produced.

Lemma 8.5. *For every smooth projective complex variety Y, with the operators of Section 2,*

$$[L^Y_{-1}, L^Y_1] = -2L^Y_0, \qquad [L^Y_0, L^Y_k] = -kL^Y_k \quad (k \geq -1,\ k \neq 0). \tag{78}$$

Proof. The classical identities

$$[\ell_{-1,Y}, \ell_{1,Y}] = 2\ell_{0,Y}, \qquad [\ell_{0,Y}, \ell_{k,Y}] = k\ell_{k,Y}$$

follow from $[\mu_Y, R_Y] = R_Y$ and $[\ell_{0,Y}, z] = z$. Normal ordering in our Hamiltonian sign convention flips the commutator sign and adds a contraction scalar between the two opposite off-diagonal polarization blocks. We compute both cases.

For the first commutator, z^{-1} crosses from nonnegative to negative polarization only at mode zero. The nonnegative output of $\ell_{1,Y}$ applied to the mode $(-z)^{-1}a$ is $(1/4 - \mu_Y^2)a$ in mode zero (the R_Y-terms still have negative powers at that output). Double-contracting the mode-zero quadratic terms in the two orders gives

$$-\frac{1}{2}\operatorname{str}(1/4 - \mu_Y^2) = -\frac{\chi(Y)}{8} + \frac{1}{2}\operatorname{str}(\mu_Y^2) = -2C_Y.$$

The sign is checked on Darboux summands: on an even coordinate $[q^2/(2\hbar), \hbar m\partial_q^2/2] = -mq\partial_q - m/2$ (scalar $-m/2$ per even line); on an odd paired plane $[\theta\psi/\hbar, -\hbar m\partial_\psi\partial_\theta] = m(1 - \theta\partial_\theta - \psi\partial_\psi)$ (scalar $+m$ per odd pair). This is exactly $-\frac{1}{2}\operatorname{str}(M)$ for $M = 1/4 - \mu_Y^2$. Thus $[L^Y_{-1}, L^Y_1] = -2(\operatorname{op}(\ell_{0,Y}) + C_Y) = -2L^Y_0$.

For the second commutator with $k > 0$, the only crossing of $\ell_{0,Y}$ is its R_Y/z part at mode zero. The opposite crossing of $\ell_{k,Y}$ uses $k-1$ factors of R_Y (in $z^{-1}(z\ell_{0,Y})^{k+1}$, a term raising loop power by one needs exactly this many zero-power R_Y factors). Its contraction with the

first crossing raises first Hodge degree by k; such endomorphisms have zero trace and zero supertrace. For $k = -1$ both crossings have the same direction, so no contraction. Thus the second commutator has no scalar. Translating $q = t - z1_Y$ preserves all commutators. □

Corollary 8.6. *Projective satisfaction of all ordinary modes by Z_Y implies $\mathsf{V}(Y)$ with exactly the constant C_Y in L_0^Y.*

Proof. Projective satisfaction says $L_k^Y Z_Y = c_k Z_Y$ with c_k independent of insertion variables. The operators differentiate neither Novikov nor $\hbar$, so they commute with every c_j. Hence their commutators annihilate Z_Y. By Lemma 8.5, $-2L_0^Y Z_Y = 0$ first; then $-kL_k^Y Z_Y = 0$ in the remaining range. □

8.5 End of the proof of the main theorem

The hypothesis $\mathsf{V}(B)$ gives projective satisfaction on B; Proposition 8.4 transports it to X; Corollary 8.6 finishes with the prescribed constant.

What was preserved. All constructions used full cohomology with categorical inverse pairings. Curve splittings, the shift correspondence, and the final Taylor expansion preserved actual integral homology labels. The conclusion therefore holds for every genus and curve class, with arbitrary ordinary descendant insertions $b_i \psi_i^{a_i}$ for $b_i \in H^*(X; \mathbb{C})$ including primitive and odd classes. Iteration over a tower of projective bundles is immediate.

Sanity check (a small example). Take $B = \mathrm{pt}$, $r \geq 1$, so $X = \mathbb{P}^r$. The state space $H^*(\mathrm{pt})$ is one-dimensional, so $\mathsf{V}(\mathrm{pt})$ reduces to a trivial one-variable computation and holds; the theorem then produces $\mathsf{V}(\mathbb{P}^r)$, matching the known toric case. Iterating covers V for towers of projective bundles over projective spaces, over target curves, and over any base known to satisfy V, without semisimplicity or splitness assumptions.

Dependencies

Used without proof.

- Virtual localization (Graber–Pandharipande): Proposition 4.1.
- Iritani's equivariant shift theorem [11]: Proposition 5.2, Lemma 5.3, Lemma 5.5.
- Quantum Riemann–Roch (Coates–Givental QRR): Lemma 7.3.
- Kontsevich–Manin, Getzler [13], and Givental's quantization formalism: Lemma 4.3 and the upper-triangular quantization in Proposition 4.1.

Related results. Eguchi–Hori–Xiong [4] and Eguchi–Jinzenji–Xiong [5] proposed the Virasoro constraints, with the Getzler [7] account of the first Hodge grading. Liu–Tian [15] prove genus zero. Givental [8, 9] and Teleman [17] prove all-genus for semisimple targets. Okounkov–Pandharipande [16] prove target curves. Coates–Givental–Tseng [3] prove toric-bundle transfer via Brown's mirror theorem [1]. Fan [6] proves all-genus reconstruction and Chern-class dependence for arbitrary projective bundles (used here only as background geometry, via the master

space W). Iritani–Koto [12] construct a mirror theorem and a D-module decomposition for projective bundles; their generic semisimplicity transfer, combined with Givental–Teleman, already gives the main conclusion when B has generically semisimple quantum cohomology. Koto [14] covers nonsplit toric bundles. The present paper makes no semisimplicity or splitness assumption.

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